

BT401 - FINAL TERM SUBJECTIVE MEGA FILE

ALL DATA FROM PAST PAPERS

GROUP: BIOTECH BRAINY BUNCH

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Current paper – 2022

1) Write the issues consideration of category VI?

Issues for consideration

- Protection of natural ecosystems and promotion of sustainable use must be integrated and mutually beneficial.
- New skills and tools need to be developed by management authorities to address the new challenges that emerge from planning, monitoring and managing sustainable use areas.
- There is also need for development of appropriate forms of governance suitable for category VI protected areas and the multiple stakeholders that are often involved.

2) Explain the threatened factors facing the forest genetic resources?

Threats

- Forests are under threat, chiefly as a result of human activities, including climate change.
- On a continent-wide scale, changes in rainfall patterns and temperature mean that some tree species will be unable to survive in their current locations. Others may be able to colonize new areas.
- Changes in climate also make it possible for pests and diseases to invade new areas, destroying the forests there.
- Overall, we can be sure that the composition and distribution of forests will change. It is because humans – and indeed the whole planet – derive so many benefits from forests and trees that we need to be concerned about how they will adapt to climate change.

3) What happen if the biodiversity of invertebrates is finished?

Because of the many ecosystem services that they provide, invertebrates have a key role to play in adapting agriculture to the effects of climate change. The extent to which the individual services provided by invertebrates will be enhanced or impeded by climate change is difficult to predict. However, if invertebrate biodiversity is lost, the capacity of ecosystems to adapt is likely to diminish. Healthy soils – and healthy, diverse soil invertebrate communities – will be vital to climate change adaptation. For example, Earthworms help to maintain soil structure and the availability of water throughout the soil profile. Studies have shown that the presence of these animals can help to alleviate the effects of drought on crop Production. Studies have also revealed the remarkable. Ability of diverse soil invertebrate communities to restore the structure of degraded soil.

4) Difference b/w law and legislation?

Legislation

The process of making or enacting laws.

OR

The preparing and enacting of laws by local state or national legislatures. In other words, it is sometimes used to apply to municipal ordinances and to the rules and regulations of administrative agencies passed in the exercise of delegated legislative functions. (INTERNET).

Laws

The system of rules which a particular country or community recognizes as regulating the actions of its members and which it may enforce by the impossible penalties.

- A person could be held guilty if he breaches Law

5) Define wild plant resources?

“Wild plant resources refers to those that grow spontaneously in self-maintaining populations in natural or semi-natural ecosystems and can exist independently.”

Wild species used by humans occur in all biogeographical regions and habitat types.

Some of the common wild plant genetic resources are as follow;

- Prickly Acacia/Keekar
- Coral Tree
- Deodar Cedar
- Dalbergia Sissoo /Sheesham Tree
- Calotropis procera/Giant milkweed
- Alovera

- Marijuana

6) What is legislation?

Legislation

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The preparing and enacting of laws by local state or national legislatures. In other words, it is sometimes used to apply to municipal ordinances and to the rules and regulations of administrative agencies passed in the exercise of delegated legislative functions. (INTERNET).

7) Who establish the Bahawalpur zoo?

Bahawalpur Zoo

Nawab of former state Bahawalpur, Sir Sadiq Muhammad Khan Abbasi, established Bahawalpur Zoo in 1942. It covers an area of 25 acres. The Bahawalpur Zoo came under administration of Wildlife Department in 1982. The lion, black tiger, fish, watch, crocodile and many other animals which were mummified during 1942 and 1974 are kept in museum. Domestic cats, jackals, an Indian civet cat, Crocodiles, lions, tigers and hyenas etc are present in zoo.

8) Explain the history of Ramsar convention?

Ramsar Convention History:

Ramsar is one of the global inter-governmental environmental agreements. The treaty was negotiated in 1960s by countries and NGOs. To avoid the increasing loss and degradation of wetland habitat for migratory water birds. In an 18 nations meeting it was adopted in the Iranian city of Ramsar. On 2nd February 1971. Came into force in 21st December 1975.

9) What is the different variations of genetic variation?

Genetic Diversity:

“The variation in the amount of genetic information within and among individuals of a population, a species, an assemblage, or a community.”

10) Daily migration?

Daily migration

Many birds make daily movements from their nest in response to environmental forces such as light, darkness temperature, humidity etc. Daily migration from their resting site to feeding area. Eg: crows, sparrows, starling.

11) Cryopreservation

Cryo-preservation

Cryo is Greek word. (krayos – frost). It literally means preservation in “frozen state.” Cryo-preservation or cryo-conservation is a process where organelles, cells, tissues, extracellular matrix, organs or any other biological constructs susceptible to damage caused by unregulated chemical kinetics are preserved by cooling to very low temperatures (typically -80 °C using solid carbon dioxide or -196 °C using liquid nitrogen).

12) Category vi

Protected Area with Sustainable Use of Natural Resources

Definition: Protected areas that conserve ecosystems and habitats, together with associated cultural values and traditional natural resource management systems. Generally large, with most of the area in a natural condition. Where a proportion is under sustainable natural resource management.

Primary objective

- To protect natural ecosystems.
- Use natural resources sustainably, when conservation and sustainable use can be mutually beneficial.

Other objectives

- To promote sustainable use of natural resources, considering ecological, economic and social dimensions
- To promote social and economic benefits to local communities where relevant;
- To facilitate inter-generational security for local communities' livelihoods – therefore ensuring that such livelihoods are sustainable;
- To integrate other cultural approaches, belief systems and world-views within a range of social and economic approaches to nature conservation;
- To contribute to developing and/or maintaining a more balanced relationship between humans and the rest of nature

13) Genetic resources of forests

Forest genetic resources

Forest genetic resources (FGR) are the heritable materials maintained within and among tree and other woody plant species that are of actual or potential economic, environmental, scientific or societal value.

14) What is cold storage?

Cold storage

It involves storage in low and non-freezing temperature.

15) Partial migration?

Partial migration

All the birds of migratory bird do not leave the native land and hence are always represented by certain individuals E.g.: finch, titmouse.

16) Cross breeding?

Crossbreeding

Crossbreeding is the mating of two animals of different breeds. Superior traits that results in the crossbred progeny from crossbreeding are called hybrid vigor or heterosis.

17) What is invasive species?

Invasive species

Alien invasive species have had severe impacts on local aquatic flora and fauna, and can upset the natural balance of an ecosystem. For example, the introduction of Nile perch to Lake Victoria has pushed many of the lake's native cichlid species to extinction.

18) Role of cryoprotectant?

A cryoprotectant is a substance used to protect biological tissue from freezing damage. Due to ice crystal formation cell membrane and cell integrity can ruptured. cryoprotectants are usually antifreeze compound.

Types of Cryoprotectants:

- 1. Membrane permitting** which can freely diffuse the membrane such as glycerol (G), ethylene glycol (EG) and dimethyl sulfoxide (DMSO)
- 2. Non Membrane permitting** which cannot permeate the cell membrane such as sugars.

19) DNA Banks?

DNA Banks

DNA banks can now be considered as a means of complimentary conservation. DNA storage is particularly useful for those species that cannot be conserved in traditional seed or field gene banks and nor conserved in situ due to high risk in that area.

Advantages

DNA banking is an efficient, simple and long-term method to conserve the genetic information.

Disadvantages

There are problems with subsequent gene isolation, cloning and transfer of DNA back to a plant and it currently does not allow the regeneration of the same genotype as the original sample.

20) Germplasm conservation of genetic resources?

CONSERVATION OF PLANT GENETIC RESOURCES

Germplasm or plant genetic resource has to be maintained in such a state that minimize the risk of loss and allows either is direct planting in the field or its preparation of planting with relative ease this is called plant genetic resources conservation or germplasm conservation.

21) Nagoya protocol?

Nagoya Protocol

The Nagoya Protocol focuses on the equitable sharing of genetic material (plant, animal, microbial, other) including the traditional knowledge associated with the genetic resources, and the benefits that arise from their use.

Component

- Access to Genetic resources
- sharing of benefits

22) Cartagena protocol on biosafety?

The Bio safety (Protocol to CBD) deals with;

- Safe handling,
- Storage
- Trans-boundary movement of the Genetically Modified Organisms (GMO).

Cartagena protocol was adopted on June 2001 in Cartagena, Spain. It entered into force on September 11th; 2003. Pakistan signed the Cartagena protocol in June 200. Pakistan has ratified it in May 2009/

23) Examples of gene flow in humans?

- Blue-eyed people from Sweden move to a small town in Mexico where people all have brown eyes. When they mate, some of their children now have blue eyes.
- A bunch of women from West Africa, where malarial is present, mate with a group of Europeans. Their children are less susceptible to contracting malaria due to the presence of antibodies from their West African mothers.

24) National monument features?

Distinguishing features

Category III protected areas are usually relatively small sites that focus on one or more prominent natural features and the associated ecology, rather than on a broader ecosystem.

25) Successful quarantine principles?

The principles of successful quarantine

One recent study (Plucknett and Smith, 1988) describes six principles of successful quarantine. They are summarized as follows

1. Sound scientific and technical principles should form the foundation of quarantine program. Pests and pathogens should be ranked by quarantine services according to the potential danger they pose to crops and the potential for success in excluding them. For example, germplasm from centres of diversity should receive a high priority because of the potential for such accessions to harbour coevolved pests or pathogens.

2. Animal and plant quarantine regulations are similar in that they may:

Require import permits issued by the quarantine service of the importing country (these may require the exporting country to certify that specified conditions have been met prior to shipment);

- Specify things that are prohibited from entry;
- Grant exceptions to the prohibitions for scientific purposes;
- Require inspection of imported materials upon arrival;
- Require appropriate treatment, if warranted, as a condition of entry; and
- Require, after arrival, quarantine or isolation in an approved facility.

3. When germplasm must be planted and grown for the purposes of quarantine testing, it should be done in an area geographically and ecologically separated from the major growing areas for that crop, to prevent the establishment of crop-specific pests or pathogens.

4. When germplasm is endangered or the need for particular accessions is particularly urgent, some discretion should be possible on the part of quarantine officials in allowing exceptions for controlled entry, despite existing regulations to the contrary.

5. Decentralized quarantine services are generally more efficient because they enfold a wider range of expertise in germplasm assessment.

6. Because delays in transit can be detrimental for any germplasm accessions, access to good communication and transportation services is essential for quarantine.

26) Botanical garden?

Botanical Garden

A botanical garden or botanic garden is a garden dedicated to the collection, cultivation and display of a wide range of plants labelled with their botanical names.

OR

It is a garden dedicated to the collection, cultivation and display of a wide range of plants labelled with their botanical names. It may contain specialist plant collections such as tropical plants, or other species of plants.

Types of plants in botanical gardens

1. cacti and succulent plants.
2. herb gardens.
3. greenhouses, shade houses.
4. tropical plants.
5. Medicinal Plants.
6. aromatic or textile plants
7. other exotic plants.

Research botanical gardens

- Abdul Wali Khan University Botanical Garden, Mardan
- Botanical Garden, Governor's House, Lahore
- Botanical Garden, Govt Zamindar College, Gujrat
- Danishmandan Botanic Garden, Lahore
- Botanical Garden, University of the Punjab, Quaid-e-Azam Campus, Lahore
- Faisalabad Botanical Gardens (part of Gatwala Wildlife Park), Faisalabad
- Forman Christian College Botanic Garden, Lahore
- Government College University Botanic Garden of GCU, Lahore

27) Disadvantage of gene bank?

Disadvantages

There are problems with subsequent gene isolation, cloning and transfer of DNA back to a plant and it currently does not allow the regeneration of the same genotype as the original sample.

28) Criteria of national park?

Distinguishing Features

The area should contain representative examples of major natural regions, and biological and environmental features or scenery.

It should be of sufficient size to maintain ecological processes.

The composition, structure and function of biodiversity should be to a great degree in a “natural” state.

29) Storage methods of cryopreservation?

There are various methods of storage:

1. **Cryopreservation** - generally involves storage in liquid nitrogen.
2. **Cold storage** - it involves storage in low and non-freezing temperature.
3. **Low pressure** – it involves partially reducing the atmospheric pressure of surrounding.
4. **Low oxygen storage** - it involves reducing the oxygen level but maintaining the pressure.

30) Hingol national park?

Hingol National Park

- Hingol National Park spread over an area of about 1,650 square km along the Makran Coast, Balochistan
- It is the largest of National Parks of Pakistan
- The area was for the first time declared reserved in 1988.
- Hingol is known to support threatened invertebrates in addition to a variety of bird species
- The park is an excellent habitat to wild animals including over 3000 ibexes, and 1500 Urials and more than 1200 Chinkara
- A number of resident and migratory birds are supported by this park.

31) Interplaying roles of plants in response to climate change?

Tree populations rely on three interplaying mechanisms to respond to environmental change:

1. Adaptation
2. Migration
3. Phenotypic plasticity

1. Adaptation and standing genetic variation

Genetic adaptations that make a population more suited for survival are achieved through gene frequency changes across generation. Many tree species have high genetic variability in adaptive traits and can therefore grow under a wide range of conditions. Indeed, phenotypic traits of

adaptive importance, such as drought tolerance, cold-hardiness, resistance to pests and diseases, and flowering and fruiting period, have been shown to vary across ecological and geographic gradients to an extent that may be as important as the differences observed amongst species.

2. Migration via pollen and seed movement

Pollen is known on occasions to travel very long distances, particularly in wind dispersed broadleaves and conifers, but also sometimes for animal-pollinated species. Pale ecological reconstructions of the decolonization of temperate zones during the Holocene have also suggested that seeds are capable of travelling long distances rapidly, in the range of several hundreds of meters per year. Landscape genetic approaches, macrofossil evidence and theoretical studies, however, indicate that cryptic refugia may have been overlooked, considerably reducing migration estimates. In addition, modern estimates of contemporary seed dispersal, although pointing to the existence of long distance dispersal events, generally indicate that median migration rates are in the range of a few tens of meters per year.

3. The role of phenotypic plasticity

Phenotypic plasticity is defined as the capacity of a particular genotype to express different phenotypes under different environmental conditions. The concept is often descended to populations and species, with plastic trees those with flexible morphology and physiology that grow at least reasonably well under a range of different environmental stresses without genetic change. A degree of phenotypic plasticity is found in most trees, but varies substantially amongst and within species. Even in species with very little genetic diversity, such as *Pinus pineal*, strong phenotypic plasticity is expressed for growth-related traits, which may have helped the species colonise new environments. At least in the short term, high plasticity is likely to favour tree survival under changing environmental conditions, although trade-offs between traits can be expected. Since phenotypic plasticity has a heritable basis and may be selected for under changing environments, complex interactions between traits are possible, depending on the magnitude and structure of change.

32) Link b/w climate change and Botanical garden?

Plants can alter the temperature of the Earth's atmosphere. Through the process of photosynthesis, plants use energy from the sun to draw down carbon dioxide from the atmosphere and then use it to create the carbohydrates they need to grow. Since carbon dioxide is one of the most abundant greenhouse gases, the removal of the gas from the atmosphere may temper the warming of our planet as a whole. Transpiration in plants can increase water vapour in the atmosphere, causing more precipitation and cloud cover in an area. The additional cloud cover often reinforces the cooling by blocking sunlight. Contribute to soil fertility and prevent soil erosion.

33) Strategies followed by effective conservation of AnGR?

Following strategies should be followed for effective conservation and utilization of AnGR:

- Formulating the National Livestock and wild-life Breeding Policies.
- Encouraging the Formation of Breed Associations.
- Developing Professional Human Resources.
- Strengthening Research and Development Institutions
- Developing Infrastructure for marketing International co-operation and assistance is needed
- in capacity building to remodel available livestock farms/research stations to conserve
- develop genetic resources

34) EEZ?

The exclusive economic zone comprises an area which extends either from the coast, or in federal systems from the seaward boundaries of the constituent states (3 to 12 nautical miles, in most cases) to 200 nautical miles (370 kilometres) off the coast.

35) Limitations of Cryopreservation?

Limitations

1. Seeds of many important tree species cannot adequately survive long-term storage;
 2. Seeds of many tropical hardwoods cannot survive even a year of conventional seed storage;
 3. Some genetic damage or change in gene frequencies may occur during seed storage.
- Cryogenic storage at its current level of technology offers great promise for long-term storage of true orthodox and sub-orthodox seeds. It has potential for application to recalcitrant species, but much more research and development will be required. Some advantages of cryogenic storage are:

- It can apparently extend storage life of true-orthodox species far beyond that possible in conventional storage;
- Genetic damage may be much less than what occurs in conventional storage. Disadvantages include
- Special equipment is required, which is not readily found in forestry facilities; (2) it is costly for large seeds
- Recalcitrant seeds (both classes) cannot be stored cryogenically with current technology.

36) Obstacles in Cryopreservation?

Obstacles

Upto 60% human body is composed of water. What's the issue then?

- Actually the freezing point of water is 0 degree centigrade while the cryoscopy temperature can be as low as -90 degree centigrade.
- Very expensive Technique
- Ice formation can result in the needle shaped crystals resulting in the damage to cell membrane.

- Unequal distribution or over distribution of cryoprotectants.
- Moreover, thermal gradients can induce mechanical stress due to uneven expansion or contraction in the biomaterial.
- The cooling rate required for optimal survival varies by several orders of magnitude between different cell types.
- Mass transfer limitations

37) Pillars of Ramsar?

Ramsar is one of the global inter-governmental environmental agreements. The treaty was negotiated in 1960s by countries and NGOs. To avoid the increasing loss and degradation of wetland habitat for migratory water birds. In 18 nations meeting it was adopted in the Iranian city of Ramsar on 2nd February 1971. Came into force in 21st December 1975

The “three pillars” of the convention.

The Contracting Parties (160) commit to:

- ❖ Work towards the wise use of all their wetlands.
- ❖ Designate suitable wetlands for the list of Wetlands of International Importance.
- ❖ Cooperate internationally on trans boundary wetlands, shared wetland systems and shared species.

38) Cryoprotectant applications?

In Animal Husbandry

The introduction of cryopreservation technology leads a major breakthrough in animal husbandry. Since the 1st successful cryopreservation of bull semen has been used to propagate the rare and endangered species using assisted reproduction techniques.

In fishery science

The 1st report on fish sperm cryopreservation was published by Blaxter (1953). To date milt (semen) of over 200 species of fresh water and marine fish have been cryopreserved and have been adequate for the purpose of cryobanking (10,11,12). In fish aquaculture the successful cryopreservation of gametes and embryos could offer new commercial possibilities, allowing the unlimited production of fry and potentially healthier and better conditioned fish as required

In medical science

Low temperature have been used in medicine and to prevent food spoilage since ancient time. Now- a- days it is used in fertility treatment the transport of human organs and the long- term storage of biological specimens, either for future or simply as a record of biodiversity.

Cryopreservation of oocyte

Human oocyte cryopreservation is a new technology in which a woman's eggs are extracted, frozen or stored. Egg freezing benefits two groups of women. One those who are diagnosed with a medical condition whereby the necessary treatments for cure may render them sterile or unable to produce viable eggs. The second who are delaying their childbearing for personal reasons. Eggs frozen at the age of 35 are more usable than fresh oocytes produced at age 43 years of age

Cryopreservation of testicular tissue

Cryopreservation of immature testicular tissue is a developing method to avail reproduction to young boys who need to have gonado toxic therapy

Embryo cryopreservation

Embryo cryopreservation is used most often to store good quality excess embryos resulting from an IVF treatment cycle. Embryos can be stored for a patient who elects to have her eggs fertilized with donar sperms. Pregnancies have been reported from embryos stored for 16 years

Cryopreservation of ovarian tissue

Ovarian tissue cryopreservation is considered to be an experimental technique for fertility preservation. This procedure is an option for patients who require immediate gonad toxic treatment of aggressive malignancies when there is insufficient time to allow the woman to undergo ovulation induction, oocyte retrieval and cryopreservation oocytes and/or embryos.

Cryopreservation of sperm:

Today human sperm cryopreservation is widely used to store donor and partner spermatozoa before assisted reproduction treatments to preserve spermatozoa before therapy for malignant diseases, vasectomy or surgical in fertility treatments and to ensure the recovery of a small number of spermatozoa in several male factor infertility

Cryopreservation of stem cell:

An important application of cryopreservation is in the freezing and storage of hematopoietic stem cell, which are found in the bone marrow rescue, hematopoietic stem cells are collected from a patient's bone marrow prior to treatment with high-dose chemotherapy. Following treatment, the patient's cryopreserved cells are thawed and infuse back into the body. This procedure is necessary, since high dose chemotherapy is extremely toxic to the bone marrow

Preservation of micro-biology cultures:

Bacteria and fungi can be kept short term refrigerated however, cell division and metabolism is not completely arrested and thus is not an optimal option for long term storage or to preserve cultures genetically or phenotypically as cell divisions can led to mutations

To conserve plant biodiversity

Conservation of plant biodiversity is an important issue concerning the human population worldwide. Conservation of plant biodiversity can be performed in situ and ex situ. These two methods are complementary and are not exclusive. They offer different alternatives for conservation, but selection of the appropriate strategy should be based on a number of criteria, including the biological nature of the species and the feasibility of applying the chosen methods. At present biotechnological methods have been used to conserve endangered, rare crop ornamental, medicinal and forest species for short-, medium-, and long-term.

Future of cryopreservation

Vitrification method of cryopreservation may bring new opportunities to research protocols. It is still an experimental procedure. There are two major concerns about vitrification - toxicity of high concentration of cryoprotectants used and microbial contamination of liquid nitrogen. Several IVF programs have adopted the vitrification method as the sole procedure for day-3 human embryos and for human blastocysts, with excellent survival and pregnancy rates. The challenge now is to find a protocol to successfully vitrify human oocytes for which the slow freezing method has yet to produce acceptable.

39) Cryoprotectant mechanism?

Until two decades ago the genetic resources were getting depleted owing to the continuous depredation by man. It was imperative therefore that many of the elite, economically important and endangered species are preserved to make them available when needed. Many methodologies have been devised for long term preservation of material.

Cryo-Preservation

Cryo is Greek word. (karyos – frost). It literally means preservation in “frozen state.” Cryo-preservation or cryo-conservation is a process where organelles, cells, tissues, extracellular matrix, organs or any other biological constructs susceptible to damage caused by unregulated chemical kinetics are preserved by cooling to very low temperatures (typically - 80 °C using solid carbon dioxide or -196 °C using liquid nitrogen).

Cryopreservation Can Be Done At

Over **Solid carbon dioxide** (at -79 degree) Low temperature deep freezer (at -80 degree)

In **Vapor Phase nitrogen** (at -150 degree)

In **Liquid Nitrogen** (at -196 degree).

40) Land race?

Land Races:

“Primitive cultivars which were selected and cultivated by farmers for many generations. Land races even respond- to –selection for high yield, but to certain extent. Since land races possess valuable alleles, their conservation is essential. The main drawbacks of land races are that they

are less uniform and low yielders. Land races have been collected in maize, sorghum, pearl millet and many other crops.

41) Areas of Cartagena protocol?

Cartagena Protocol areas

The Protocol covers:

- Transboundary movement, transit, handling and use of all living modified organisms that may have adverse effects on the conservation and sustainable use of biological diversity, taking into account risks to human health.

It does not cover:

- Products derived from LMOs (e.g. paper from GM trees)
- LMOs, which are pharmaceuticals for humans that are addressed by other relevant international agreements or organizations.

42) Gene pool?

The gene pool is the set of all genes, or genetic information, in any population, usually of a particular species. A gene pool is a collection of all the genes in a population. This can be any population - frogs in a pond, trees in a forest, or people in a town. A gene pool isn't a real place; it's an abstract idea or an imaginary collection. The gene pool is all the different alleles that are present in a population. For example; you have a population of goats or cows of different colours. The colours are caused by different variations of the same gene. All the different variations of the gene make up the gene pool. The gene pool does not bother with frequencies; a variant is either present or not.

43) PGR conservation?

Plant genetic resources

Plant Genetic Resources for Food and Agriculture (PGRFA) is the conservation of raw material that farmers and plant breeders use to improve the quality and productivity of crops. They can be defined as any genetic material of plant origin of actual or potential value for food and agriculture, e.g. seeds, tubers, mature plants etc.

44) Strict nature reserve?

Strict Nature Reserve:

Category Ia are strictly protected areas set aside to protect biodiversity and also possibly geological/geomorphic features, where human visitation, use and impacts are strictly controlled and limited to ensure protection of the conservation values. Such protected areas can serve as indispensable reference areas for scientific research and monitoring.

45) Cryopreservation future

Future of cryopreservation

Vitrification method of cryopreservation may bring new opportunities to research protocols. It is still an experimental procedure. There are two major concern about vitrification - toxicity of high concentration of cryoprotectants used and microbial contamination of liquid nitrogen. Several IVF programs have adopted the vitrification method as the sole procedure for day-3 human embryos and for human blastocysts, with excellent survival and pregnancy rates. The challenge now is to find a protocol to successfully vitrify human oocytes for which the slow freezing method has yet to produce acceptable.

46) Cold storage?

Cold storage

It involves storage in low and non-freezing temperature.

47) PEPA

Pakistan Environmental Protection Act, 1997

PEPA provides for;

- Protection,
- Conservation,
- Rehabilitation and
- Improvement of the environment;
- PEPA provide framework for prevention and control of pollution
- Helps in protection of sustainable development.

48) Special about agricultural genetic resources?

- To feed the world, we all need these resources.
- Agricultural resources have been shared and exchanged over thousands of years. Mostly it is impossible to identify a single country of origin.
- Countries and regions are “interdependent”: they all depend for their food and agriculture on crops that originated elsewhere.

49) Issues considered in strict nature reserve?

Issues for consideration

- There are few areas not under some kind of legal or at least traditional ownership, so that finding places that exclude human activity is often problematic.
- Most apparent problem is with climate and air pollution

- New and emerging diseases.
- In an increasingly modified ecology, it may become increasingly difficult to maintain pristine areas through non-intervention.

50) What is coevolution?

“Plant domestication is the process whereby wild plants have been evolved into crop plants through artificial selection.” This involves an early hybridization event followed by selective breeding. Domestication is the result of an immensely slow process, hundreds or thousands of years. This is called co- evolution because during domestication both plants and human behaviours evolved to suit one another.

51) National animal of Pakistan and its global population?

Alpine Markhors

This wild goat is found in the northern mountain ranges of Pakistan. Markhor is the national animal of Pakistan. The total global population of Markhors is estimated to be 2,500. About 1,500 Markhors are found in Pakistan. This species is in high risk of extinction now a days.

52) Difference between natural and artificial barriers?

Physical barriers to gene flow both “natural” and “artificial”

- Natural physical barriers include mountain ranges, oceans or vast deserts
- Artificial physical barriers are man-made barriers such as “The Great China Wall”, barrages or dams etc.

53) Storage strategy of DNA bank?

Storage strategy

Determining what to store and for how long is an important consideration, used to determine sample size, capacity of the DNA bank, preparation of samples and documentation. Long-term needs and expected volume and number of samples to be stored will determine organization and repository design.

54) Mechanism of cryopreservation techniques in plants?

The cryopreservation technique followed by the regeneration of plants involves following steps

- Selection of material.
- Addition of cryoprotectant.
- Freezing.
- Storage in liquid nitrogen.
- Thawing.

- Washing and re-culturing.
- Measurement of viability.
- Regeneration of plants.

1. Selection of plant material

Selection of proper plant material is important. Two important factors depend on it such as (a) nature and (b) density. Any tissue can be selected for this purpose. e.g. meristem, embryo, ovules, seeds etc.. The density should be high.

2. Addition of cryoprotectant

They are chemical which prevent cryo-destruction. These are sucrose, alcohols, glycols, some amino acid (proline), DMSO (dimethyl sulfoxide). Generally two Cryo protective agents should be used together instead of single one as they are more effective. Two common cryoprotective agents are dimethyl sulfoxide (DMSO) and glycerol. Glycerol is used primarily for cryoprotection of red blood cells, and DMSO is used for protection of most other cells and tissues

3. Freezing

Cryopreservation is based on the ability of certain small molecules to enter cells and prevent dehydration and formation of intracellular ice crystals, which can cause cell death and destruction of cell organelles during the freezing process. The sensitivity of cells to low temperature depends on the plant species. There are four different

Types of methods:

1. **Slow freezing method** - the tissue or plant material is slowly frozen at slow cooling rate. The advantage is the plant cells are partially dehydrated and survive better.
2. **Rapid freezing method** - it involves plunging the vials in liquid nitrogen. The temperature decreases from -300 to -1000 degree rapidly.
3. **Combined freezing method** - this is combination of both slow and rapid freezing method. The process is carried out in step wise like manner.
4. **Dry freezing method** - in this method dehydrated cells and seeds are stored.

4. Storage

The maintenance of the frozen cells or material at specific temperature is very important. In general the temperature is kept -70 to -196 degree. Prolong storage is done at temperature of -196 degree in liquid nitrogen. To prevent damage, continuous supply of nitrogen is done.

5. Thawing

Usually carried out by plunging the vials into warm water bath with vigorous swirling. As thawing occurs the vials are transferred to another bath at 0 degree.

6. Washing and re-culturing

The preserved material is washed few times to remove the cryoprotectant. This material is then recultured in a fresh medium

7. Measurement of viability

8.Plant regeneration

The viable seeds are cultured on non-specific growth medium. Suitable environmental conditions are maintained.

54) Habitat species management areas?

Habitat-Species Management Area

Definition: IUCN Management Category IV (Habitat/Species Management Area) refers to areas that are managed to protect particular species or habitats. They are defined by IUCN as “protected areas aiming to protect particular species or habitats and management reflect this priority.

Primary objective: To maintain, conserve and restore species and habitats.

Other objectives

- To protect vegetation patterns.
- To protect fragments of habitats as components of landscape or seascape-scale conservation strategies.
- To develop public education and appreciation of the species or habitat
- Developed by the International Union for Conservation of Nature (IUCN) with support of the World Commission on Protected Areas (WCPA) and other international institutions such as the Convention on Biological Diversity (CBD), and assigned by national governments.

Distinguishing features

- Protection of particular species
- Protection of habitats
- Active management to maintain target species
- Active management of culturally-defined ecosystems

Role in the landscape/seascape

- Protect critically endangered populations of species.
- Protect rare or threatened habitats.

- Provide flexible management strategies and options in buffer zones around, or connectivity conservation corridors between, more strictly protected areas that are more acceptable to local communities and other stakeholders;
- Maintain species that have become dependent on cultural landscapes where their original habitats have disappeared or been altered.

Management

These areas are managed to maintain or restore certain species and/or habitats, and this can be carried out by a range of actors depending on the governance type of the area (see IUCN Protected Area Management Categories for information on governance types). These are often areas that have already undergone substantial modification, where a high degree of human pressure often exists, and significant management intervention is necessary. Category IV areas will generally be publically accessible.

Legal and compliance

The classification of a category IV protected area requires that such areas are managed for conservation by legal or other effective means, and therefore legal recognition and protection at the national or subnational level is likely to be present in these sites. The level of legal protection will however vary between countries, and will depend on the governance type of the area, as they receive differing levels of recognition by government in different countries. Nonetheless a number of national laws are likely to apply to these sites that deter large-scale economic activities in order to maintain the conservation values of these areas.

Biodiversity importance

Category IV areas are important for their role in ‘plugging the gaps’ in conservation strategies by protecting key species or habitats in ecosystems. It provides a management approach for areas that have already undergone substantial modification, necessitating protection of remaining fragments for identified target species with or without intervention.

Socio-cultural values

- While Category IV areas are not necessarily associated with human presence and intervention, many exist in crowded landscapes and seascapes where human pressure is comparatively greater, both in terms of potential illegal use and visitor pressure
- Many category IV protected areas exist in crowded landscapes and seascapes.
- The category IV protected areas that rely on regular management intervention need appropriate resources from the management authority.
- Because they usually protect part of an ecosystem, successful long-term management of category IV protected areas necessitates careful monitoring and an even greater-than-usual emphasis on overall ecosystem approaches and compatible management in other parts of the landscape or seascape.

55) Cholistan wildlife sanctuary?

Cholistan Wildlife Sanctuary

Cholistan Wildlife Sanctuary is part of the Cholistan desert in the south eastern portion of the province of Punjab and contains some of the most rare and interesting wildlife in Pakistan. Some of the rare animals of this region are Desert wolf (rare), Indian fox, Red fox, Jackal, Small Indian civet, Small Indian mongoose, Indian grey mongoose, Indian desert cat, Jungle cat, Caracal cat, Chinkara gazelle, Blackbuck, Nilgai antelope, Houbara bustard, Peregrine falcon, Saker falcon, Black backed vulture, Indian cobra, Monitor lizard, Saw scaled viper and Russells viper.

56) Advantage of tissue culture technique?

Advantages of tissue culture conservation

- Source of disease-free material.
- It is most appropriate for rapid multiplication purposes, dissemination and active collections.

57) Treaty?

“Treaty, a binding formal agreement, contract, or other written instrument that establishes obligations between two or more subjects of international law”

- Treaties do not need to follow any special form.
- A treaty often takes the form of a contract, but it may be a joint declaration or an exchange of notes.

Pakistan is signatory to a large number of international treaties and conventions. These are listed below;

- Convention on biological diversity, 1992
- Cartagena protocol on Bio Safety, 2001
- CITES, 1973
- Ramsar Convention, 1971
- Convention on the conservation of migratory species, 1981

58) Landscape seascape?

Definition:

A protected area where the interaction of people and nature over time has produced an area of distinct character with significant ecological, biological, cultural and scenic value.

Primary objective:

To protect and sustain important landscapes/seascapes and the associated nature conservation.

Other objectives

- To maintain a balanced interaction of nature and culture.
- To contribute to broad-scale conservation by maintaining species associated with cultural landscapes and/or by providing conservation opportunities in heavily used landscapes;
- To provide opportunities for enjoyment, well-being and socio-economic activity through recreation and tourism;
- To provide natural products and environmental services;
- To provide a framework to underpin active involvement by the community in the management of valued landscapes or seascapes and the natural and cultural heritage that they contain;
- To encourage the conservation of agrobiodiversity and aquatic biodiversity;
- To act as models of sustainability so that lessons can be learnt for wider application.

59) Responsible for botanical garden?

Botanical gardens are often run by universities or other scientific research organizations, and often have associated herbaria and research programmers in plant taxonomy or some other aspect of botanical science.

60) Cartagena protocol?

The Bio safety (Protocol to CBD) deals with;

- Safe handling,
- Storage
- Trans-boundary movement of the Genetically Modified Organisms (GMO).

Cartagena protocol was adopted on June 2001 in Cartagena, Spain. It entered into force on September 11th; 2003. Pakistan signed the Cartagena protocol in June 200. Pakistan has ratified it in May 2009.

61) Impact of climate change on FGR?

Climate change may also result in high variability in temperature and precipitation, with an increase in incidence of extreme events, such as flooding, late frosts and intensive summer droughts, amongst other events. In some areas, such as the Mediterranean and the Neo-tropics, an increase in seasonality is also expected. Under such conditions, natural selection may not result in efficient adaptation because selection pressures are multi-directional, involving traits that may be inversely correlated at the gene level. The standing genetic variation in populations may then not be large enough to create the rare new genotypic combinations that are required. Ecosystems affected by abrupt change may sustain rapid and widespread transformation as ecological tipping points are exceeded. Given the pivotal role of trees in ecosystem function, abrupt climate change impacts on them may thus have profound consequences for forests as a

whole. Irreversible loss of ecosystem integrity and function may follow, with replacement by new no endemic ecosystems.

Direct impacts of climate change

These include high tree mortality through extreme climatic events, particularly drought in combination with widespread regeneration failure, for example, examined the evidence for anthropogenic climate change leading to future large-scale “dieback” in Amazonian rain forest. Analysis suggested that dry season water stress is likely to increase in eastern Amazonia over the 21st century, with the region tending toward a climate more appropriate to seasonal forests.

Effects of changing climate on organisms associated with trees

In particular, changes in the biology of insect pests and diseases may make ecosystems more susceptible to tree mortality. Because of improved environmental conditions for the pest and reduced tree resistance due to increased stress, pests may react to climate change with range expansions and/or increases in attack severity.

Changes in abiotic disturbance regimes

These include changes in fire regimes, flooding, landslides and/or hurricanes. Fire and climate are closely linked and are also associated with changes in land use. Coupled climate and fire-risk models suggest not only an increase in the frequency of fires but also in fire size and length of the fire-risk season, with some areas subject to risk that were not before. Malhi et al. (2009) considered how tipping points may be reached in Amazonian rainforest by a combination of increased dryness and an increased incidence of fire events

Invasion by organisms foreign to local ecosystems

This includes the increased risk of establishment by invasive species which accidentally arrive into ports of entry, through globalized commerce. By making new niches available, climate change will facilitate the survival of mammals, insects, diseases and/or weeds foreign to endemic ecosystems.

62) When Black African Rhinoceros is declared as extinct?

West African Black Rhinoceros

The West African black rhinoceros (*Diceros bicornis longipes*) was a subspecies of the black rhino that was declared extinct in 2011.

63) Why mostly species are facing threat in Pakistan?

Overexploitation of natural sources – overhunting, overfishing and forest logging – threatens the largest number of species. Agriculture – the expansion of land for crops, livestock, timber and aquaculture – is another leading threat. This is driven by habitat loss, livestock competition and agricultural pollution. The result: deforestation, overgrazing, soil erosion, salinity and waterlogging, non-sustainable agricultural practices and hunting have become major threats to biodiversity.

64) Purposes of zoological areas?

The zoo originally evolved from the menageries of the ancient world, in which royalty would exhibit their collection of exotic pets. Unfortunately not all zoos are scientific in nature, and extreme controversy has arisen regarding how the animals are treated. Suffice to say, regulation is necessary to ensure proper care. Conservation (not exploitation) should always be the central goal behind any legitimate zoo. Zoos provide the opportunity for people to see a glimpse of this side of nature. Zoo plays important role in conservation of many threatened/endangered species. Wild animals in captivity help us manage and conserve them in the wild. Zoos raise money for conservation efforts. Zoo provides area for recreational purposes

65) Define forest genetic resources?

Forest genetic resources

Forest genetic resources (FGR) are the heritable materials maintained within and among tree and other woody plant species that are of actual or potential economic, environmental, scientific or societal value.

66) Name of Ramser in Punjab?

Punjab

- Uchhali Complex
- Taunsa Barrage
- Chashma Barrage

67) Application of conserved germplasm?

Storage of seed is indispensable to most of plantation forestry, and the practice should not be dismissed too readily as a basic tool in maintaining genetic diversity. Conventional seed storage offers several advantages:

1. Seeds of many valuable species can survive long-term storage ('longterm' is defined as spanning a period of time longer than one rotation)
2. Good storage facilities are now available in most of the world, and they are used extensively for tree seed storage for various regeneration purposes;
3. Seed storage is a relatively cheap method for conserving a broad range of germplasm
4. Large land areas are not tied up in conservation
5. International exchange of genetic material is facilitated by seed storage.

68) Clonal Repository?

Clonal repository is field Gene bank where genetic resources of clonally propagated crops like fruits are preserved as living plants. Various institutions are involved in the capacity building to develop AnGR, in the country.

69) Describe cryopreservation with freezing methods?

Freezing

Cryopreservation is based on the ability of certain small molecules to enter cells and prevent dehydration and formation of intracellular ice crystals, which can cause cell death and destruction of cell organelles during the freezing process.

The sensitivity of cells to low temperature depends on the plant species. There are four different types of methods:

1. Slow freezing method - the tissue or plant material is slowly frozen at slow cooling rate. The advantage is the plant cells are partially dehydrated and survive better.

2. Rapid freezing method - it involves plunging the vials in liquid nitrogen. The temperature decreases from -300 to -1000 degree rapidly.

3. Combined freezing method - this is combination of both slow and rapid freezing method. The process is carried out in step wise like manner.

4. Dry freezing method - in this method dehydrated cells and seeds are stored.

70) Bottleneck effect?

Bottleneck effect

The bottleneck effect is a sharp lowering of a population's gene pool because of an environmental, or human-caused, change. It is an extreme example of genetic drift that happens when the size of a population is severely reduced. Events like natural disasters (earthquakes, floods, fires) can decimate a population, killing most individuals and leaving behind a small, random assortment of survivors.

Bottleneck effect and Allele Frequencies:

Allele frequencies in a group may be very different from those of the population prior to the event, even some alleles may be missing entirely. The smaller population will also be more susceptible to the effects of genetic drift for generations (until its numbers return to normal). Effect potentially causing even more alleles to be lost.

Examples

In human evolution

It is theorized, based on genetic evidence, that a few tens of thousands of years ago the population of Homo sapiens was reduced for a period to a few thousand or tens of thousands of

people. Such a bottleneck would explain the extremely low level of genetic diversity found within our species, when contrasted with others, such as Chimps.

Cheetah

All Cheetah shared a small number of alleles.

- Less than 1% diversity
- As if all cheetahs are identical twins

Bottleneck effect

- 10,000 years ago
- Ice age
- Last 100 years
- Poaching and loss of habitat

71) In-vivo gene bank?

In vivo Gene Bank:

Generally plant seeds, vegetative propagules are used for storage for long time. The whole plants are preserved. This type of conservation strategy is called in vivo gene bank. In this approach, conservation method of storage is used for preservation of plant genetic resources.

72) Date and place where Nagoya protocol is adopted?

The protocol was adopted on 29 October 2010 in Nagoya, Japan, and entered into force on 12 October 2014. It has been ratified by 97 parties, which includes 96 UN member states and the European Union. It is the second protocol to the CBD; the first is the 2000 Cartagena Protocol on Biosafety.

73) Which type of material is used in DNA bank?

Most DNA provided by DNA banks is used for studies to attempt to develop more productive or more environmentally friendly agricultural species. Some DNA banks also store the DNA of rare or endangered species to ensure their survival. The DNA bank can be used to compare and analyse DNA samples.

74) Invasive species?

Invasive species

Alien invasive species have had severe impacts on local aquatic flora and fauna, and can upset the natural balance of an ecosystem. For example, the introduction of Nile perch to Lake Victoria has pushed many of the lake's native cichlid species to extinction.

75) Effect of humans on migration of invertebrates?

Human activities have created barriers to the migration of invertebrate species. These barriers are likely to affect species in natural ecosystems rather more severely than those associated with agro-ecosystems. The movement of the latter is likely to be facilitated rather than hindered by human-induced landscape changes. In situ adaptation of invertebrate species is expected to be most marked where movement is not an option (e.g. on low, isolated islands).. It is very difficult to predict how the combined effects of changing temperatures, changing rainfall patterns and elevated carbon dioxide levels will affect invertebrates and their capacities to provide ecosystem services or to act as pests. As yet, few studies have attempted to investigate interactions of this kind. Further complexity is added by the prospect that the other components of the ecosystem with which invertebrates interact – food plants, micro-organisms, etc. will also be affected by climate change.

76) Criteria of National Parks?

Distinguishing Features

The area should contain representative examples of major natural regions, and biological and environmental features or scenery. It should be of sufficient size to maintain ecological processes. The composition, structure and function of biodiversity should be to a great degree in a “natural” state.

77) Socio-culture values of National monument?

Category III areas are likely to hold socio-cultural values as they may have features such as sacred groves, springs, waterfalls, mountains, sea coves etc. of importance to one or more faith groups. These areas are often of significant tourism value and can be managed with the objective of promoting sustainable tourism.

78) Features of West Pakistan Fisheries act, 1961?

West Pakistan Fisheries Ordinance, 1961

- It is an ordinance to amend and consolidate the law relating to fisheries in the West Pakistan.
- It says that no person will be allowed to use dynamite, pesticides or other explosives for catching the fish.
- Government can declare any water area as sanctuary.
- Then there will be no permit for fishing except some license.
- An Inspector of Fisheries may without a warrant arrest any person committing
- In his view any offence under section 6, 7, 8, 9 or 11

79) What is assistant migration?

Assisted population migration (also assisted genetic migration or assisted gene flow) – moving seed sources or populations to new locations within the historical species range.

80) Names of labs present in Pakistan?

Plant Genetic Resources Institute hosts the sole National Gene bank of Pakistan for conservation of plant genetic resources and six labs including

- germplasm exploration lab
- seed preservation lab
- in vitro conservation lab
- germplasm evaluation lab
- plant introduction and seed health lab
- Data management lab

81) Write two advantages of cryopreservation?

Advantages:

- Cryopreservation helps in the preservation of biological materials.
- Cryopreservation is used to maintain the biosynthetic properties of plants
- Sperm, gametes, embryos, tissues, bone marrow, and organ can be preserved.
- Helps to study the adapting nature of plants and animals under the low temperature.
- Used to preserve the genetic materials of the plants which are on the verge of extinction
- Prevent in breeding
- Reduced risk of microbial contamination
- Reduced risk of cross contamination with other cell lines
- Reduced risk of genetic drift and morphological changes
- Embryo cryopreservation is used most often to store good quality excess embryos resulting from an IVF treatment cycle.
- Embryos can be stored for a patient who elects to have her eggs fertilized with donor sperms. Pregnancies have been reported from embryos stored for 16 years.
- Human sperm cryopreservation is widely used to store donor and partner spermatozoa to preserve sperms
- It also ensure the recovery of a small number of spermatozoa in several male factor infertility¹⁷.
- It is commonly called sperm-banking is a procedure to preserve sperm cells.

82) Write role of national park in land scape or sea scape?

Role in the landscape/seascape

- Protecting larger-scale ecological processes.
- Protecting compatible ecosystem services.
- Protecting particular species and communities that require relatively large areas of undisturbed habitat.
- To inform and excite visitors about the need for and potential of conservation programmes.
- To support compatible economic development, mostly through recreation and tourism, that can contribute to local and national economies and in particular to local communities.

83) Link between forest genetic resources (FGR) and climate change?

Forest Genetic Resources and Climate Change

Diversity of forest genetic resources enables the potential for a species (or a population) to adapt to climatic changes and related future challenges such as temperature changes, drought, pests, diseases and forest fires. Though forest trees are known for showing great plasticity in their response to climate changes, not all species are naturally capable to adapt at the pace necessary.

Climate change may also result in high variability in temperature and precipitation, with an increase in incidence of extreme events, such as flooding, late frosts and intensive summer droughts, amongst other events. In some areas, such as the Mediterranean and the Neo-tropics, an increase in seasonality is also expected. Under such conditions, natural selection may not result in efficient adaptation because selection pressures are multi-directional, involving traits that may be inversely correlated at the gene level. The standing genetic variation in populations may then not be large enough to create the rare new genotypic combinations that are required. Ecosystems affected by abrupt change may sustain rapid and widespread transformation as ecological tipping points are exceeded.

Given the pivotal role of trees in ecosystem function, abrupt climate change impacts on them may thus have profound consequences for forests as a whole. Irreversible loss of ecosystem integrity and function may follow, with replacement by new non endemic ecosystems.

84) Background history of Ramsar conservation?

History: Ramsar is one of the global inter-governmental environmental agreements. The treaty was negotiated in 1960s by countries and NGOs. To avoid the increasing loss and degradation of wetland habitat for migratory water birds. In an 18 nations meeting it was adopted in the Iranian city of Ramsar On 2nd February 1971. Came into force in 21st December 1975.

85) Write strategies and utilization of AnGR?

Following strategies should be followed for effective conservation and utilization of

AnGR:

- Formulating the National Livestock and wild-life Breeding Policies.
- Encouraging the Formation of Breed Associations.

- Developing Professional Human Resources.
- Strengthening Research and Development Institutions
- Developing Infrastructure for marketing International co-operation and assistance is needed
- in capacity building to remodel available livestock farms/research stations to conserve and
- develop genetic resources

86) Why do we protect endangered species?

“A species of plant or animal that is in immediate danger of becoming extinct and needs protection to survive.” Endangered species are like fire alarms. They tell us about problems in our home we call Earth. Endangered species must be protected for better health of earth and ecosystem balancing. Pakistan is host to a number of endemic species and sub species. Some birds have their principal surviving population located in Pakistan. The Northern Areas of Pakistan provide a habitat for a number of globally significant wildlife species. The survival of these species is threatened by loss of habitat stemming from human activities, excessive hunting by locals and outsiders, and weak enforcement of wildlife protection law.

87) Process of Sampling in DNA?

Processing of samples

DNA preserved in DNA banks will be stored either within cells and extracted upon retrieval from storage or extracted from cells and purified before storage. The quality of the DNA is expressed through yield, purity, molecular weight, amplification efficiency and authenticity of sequences. The quality of DNA extracted from plant specimens is dependent on the condition of the specimen before storage, the storage environment and the duration of storage. Rapid drying of plant samples with silica gel or lyophilisation helps to preserve the DNA.

88) Passenger pigeon history and prevalence?

Passenger Pigeon

The passenger pigeon may have once constituted 25 to 40 percent of the bird population. The 19th century brought widespread hunting and trapping of the birds, which severely diminished their populations. The last passenger pigeon, named “Martha” died at age 29 at the Cincinnati Zoo in 1914.

Prevalence

Prevalence refers to the total number of individuals in a population who have a disease or health condition at a specific period of time, usually expressed as a percentage of the population.

89) If invertebrates eliminate from earth what happen?

Perhaps the most neglected group of all – in research, in farming practices, and in policies and strategies for agriculture and biodiversity – are the soil-dwelling invertebrates

- Small, out-of-sight and uncharismatic these animals may be, but their significance is enormous. Some larger soil-dwelling invertebrates, such as earthworms, ants and termites, have been described as “ecosystem engineers”. They create the physical structures needed to maintain healthy soil communities and for basic soil processes such as water infiltration and storage, and sequestration and cycling of carbon.
- They help maintain the chemical fertility needed for plant growth. Also vitally important are the invertebrates that process the leaf litter that falls onto the surface of the soil

90) Wildlife sanctuary?

“Sanctuary is an area which is of adequate ecological, faunal, floral, Geo-morphological, natural or zoological significance.”

The Sanctuary is declared for the purpose of protecting, propagating or developing wildlife or its environment.

Wildlife Sanctuaries of Pakistan are;

- Astor Wildlife Sanctuary
- Baltistan Wildlife Sanctuary
- Chasma and Taunsa Barrage Dolphin Sanctuary
- Cholistan Wildlife Sanctuary
- Hab Dam Wildlife Sanctuary
- Kargah Wildlife Sanctuary
- Mahal Kohistan Wildlife Sanctuary
- Naltar Wildlife Sanctuary
- Nara Desert wildlife Sanctuary
- Rann of Kutch Wildlife Sanctuary

91) Some advantages of tissue culture technique?

Advantages of tissue culture conservation

- Source of disease-free material.
- It is most appropriate for rapid multiplication purposes, dissemination and active collections.

92) Name of wetlands under ramsar conservation?

Religion wise significant ramsar sites of Pakistan

Baluchistan

- Astola Island
- Hub Dam
- Jiwani Coastal Wetland
- Miani Hor
- Ormara Turtle beach

Sindh

- Keenjhar (Kalri) Lake
- Haleji Lake
- Drigh Lake
- Indus Dolphin reserve
- Jubho lagoon
- Nurri Lagoon
- Deh Akro-II
- Indus Delta
- Runn of Kutch

Punjab

- Uchhali Complex
- Taunsa Barrage
- Chashma Barrage

Khyber Pakhtunkhwa

- Tanda Dam
- Thanedar Wala

93) Great migration?

The Great Migration

The Great Migration isn't called great by chance. Every year in the Great Rift Valley of Tanzania and Kenya, which are located in Africa, more than 1.5 million wildebeest migrate northwest across the grassy plains of the Serengeti. Zebras and other grazing animals also take part in this migration. These animals are constantly on the search for food, following the grasses that flourish during the rainy season.

94) How to conserve plant genetic resources?

Conservation of plant genetic resources

- Selection of target taxa
- Project commission
- Eco geographic survey/preliminary survey mission
- Conservation objectives
- Field exploration
- Conservation strategies
- Conserved product deposition and dissemination
- Characterization/ Evaluation
- PGR utilization
- Utilization products

95) Phenotypic plasticity with respect to the environment?

Phenotypic plasticity is defined as the capacity of a particular genotype to express different phenotypes under different environmental conditions. The concept is often descended to populations and species, with plastic trees those with flexible morphology and physiology that grow at least reasonably well under a range of different environmental stresses without genetic change. A degree of phenotypic plasticity is found in most trees, but varies substantially amongst and within species. Even in species with very little genetic diversity, such as *Pinus pineal*, strong phenotypic plasticity is expressed for growth-related traits, which may have helped the species colonise new environments. At least in the short term, high plasticity is likely to favour tree survival under changing environmental conditions, although trade-offs between traits can be expected. Since phenotypic plasticity has a heritable basis and may be selected for under changing environments, complex interactions between traits are possible, depending on the magnitude and structure of change.

96) Vegetables ratio in Pakistan?

Pakistan covering 75% of the total area under vegetables, accounting for 74% of the total production. The major share in the production is of Punjab (63%) followed by Sindh (14%), Baluchistan (12%) and KPK (11%). Maximum area is grown under potatoes and about 88 % occurs in Punjab. About 46% of onion is cultivated in Sindh and 25% in Punjab. Chili is at the third position of which 84% is cultivated in Sindh.

97) What is antibiotics? Give an example?

Antibiotics are chemical substances produced by a living organism that kill or stop the growth of disease-causing micro-organisms such as fungi and bacteria. Examples of antibiotics: Penicillin. Streptomycin.

98) What is the role of national parks?

- Protecting larger-scale ecological processes.
- Protecting compatible ecosystem services.
- Protecting particular species and communities that require relatively large areas of undisturbed habitat.
- To inform and excite visitors about the need for and potential of conservation programmes.
- To support compatible economic development, mostly through recreation and tourism, that can contribute to local and national economies and in particular to local communities.

99) Objectives of CBD?

Objectives of CBD, 1992

The three inter-related objectives are:

- The conservation of biological diversity;
- The sustainable use of its components;
- The fair and equitable sharing of the benefits arising out of the utilization of genetic resources, including by appropriate
- Access to genetic resources,
- Transfer of relevant technologies,
- Funding.

100) Issue in Category V?

- Landscape and/or coastal and island seascape of high and/or distinct scenic quality and with significant associated habitats, flora and fauna and associated cultural features
- A balanced interaction between people and nature
- Unique or traditional land-use patterns

101) Limitations of Germplasm conservation?

Limitations

1. Seeds of many important tree species cannot adequately survive long-term storage;
2. Seeds of many tropical hardwoods cannot survive even a year of conventional seed storage;

3. Some genetic damage or change in gene frequencies may occur during seed storage. Cryogenic storage at its current level of technology offers great promise for long-term storage of true orthodox and sub-orthodox seeds. It has potential for application to recalcitrant species, but much more research and development will be required. Some advantages of cryogenic storage are:

- It can apparently extend storage life of true-orthodox species far beyond that possible in conventional storage;
- Genetic damage may be much less than what occurs in conventional storage. Disadvantages include
- Special equipment is required, which is not readily found in forestry facilities; (2) it is costly for large seeds
- Recalcitrant seeds (both classes) cannot be stored cryogenically with current technology.

102) What is Game reserve?

“A game reserve is an area wherein controlled hunting and shooting is permitted on permit basis”

- A game reserve (wildlife preserve) is a large area of land where wild animals live safely or are hunted in a controlled way for sport.
- In the game reserves the major focus is specifically the animals.
- If hunting is prohibited, a game reserve may be considered a “Nature Reserve”
- Wherein all aspects of naturally-occurring life in the area are considered.
- Most of the areas in game reserves have created to provide habitat protection for animal species commonly referred to as game (hunnable species for sport or meat).

103) How is Allopatric speciation different from sympatric speciation?

Allopatric speciation is speciation that results when a population is separated by a physical barrier. It is also referred to as geographic speciation. Sympatric speciation is speciation that occurs without physical separation of members of the population.

104) Danger to aquatic resources by aquatic pollution?

Danger to Aquatic Genetic Resources:

Aquatic pollution:

- Pollution caused by industrial effluents and wastewater of private households has severely damaged the habitats of many fish species
- Nutrient input through farming has resulted in algal blooms
- Oil tanker disasters causing oil spills destroyed entire coastal areas

- Above issues can cause a loss of general fitness of the animals, remain stunted in growth and become less fertile
- Famous example of effect of aquatic pollution is “MINAMATA Disease.” This disease was spread due to consuming mercury poisoned fishes by people

105) Conservation of Diversity of habitat species management?

Biodiversity importance

Category IV areas are important for their role in ‘plugging the gaps’ in conservation strategies by protecting key species or habitats in ecosystems. It provides a management approach for areas that have already undergone substantial modification, necessitating protection of remaining fragments for identified target species with or without intervention.

106) Phenotypic plasticity?

Phenotypic plasticity is defined as the capacity of a particular genotype to express different phenotypes under different environmental conditions.

107) Huntington disease?

Huntington’s disease

Huntington’s disease (aka Huntington’s chorea) is a genetic disorder which results in slowly progressing brain cell death. There are two distinct populations in which the disorder occurs much more often.

1. The first group is the Afrikaner population of South Africa.
2. The second group is the residents of the Lake Maracaibo region of Venezuela.

108) Three research botanical gardens in Pakistan?

- Cacti and succulent plants.
- Herb gardens
- Greenhouses, shade houses.
- Tropical plants.
- Medicinal Plants.
- Aromatic or textile plants
- Other exotic plants.

109) Application of cryopreservation in animal husbandry?

In Animal Husbandry

The introduction of cryopreservation technology leads a major breakthrough in animal husbandry .Since the 1st successful cryopreservation of bull semen has been used to propagate the rare and endangered species using assisted reproduction techniques.

110) What is the effect of extreme temperature in invertebrates?

Climate change is expected to affect all three of the main groups of invertebrate ecosystem-service providers as well as invertebrate pests. Invertebrates have limited ability to control their body temperatures. Therefore, although some groups such as soil-dwelling organisms are to some degree buffered against the effects of temperature fluctuations in the wider environment, it is likely that rising temperatures will directly influence the distribution of invertebrate species. Many of the challenges associated with the management of invertebrate genetic resources in agriculture in the context of climate change will relate to climate-driven or human-assisted movement of invertebrate species.

111) What is genetic material?

Any material of plant, animal, microbial or other origin that carries genetic information and that passes it from one generation to the next. The information contained controls reproduction, development, behaviour, etc.

112) Objectives of habitat management species?

Primary objective: To maintain, conserve and restore species and habitats.

Other objectives

- To protect vegetation patterns.
- To protect fragments of habitats as components of landscape or seascape-scale conservation strategies.
- To develop public education and appreciation of the species or habitat
- Developed by the International Union for Conservation of Nature (IUCN) with support of the World Commission on Protected Areas (WCPA) and other international institutions such as the Convention on Biological Diversity (CBD), and assigned by national governments.

113) What is a zoological garden?

“A zoological park is a location where animals are kept in captivity for study and viewing.”

Examples:

- Bahawalpur Zoo
- Changa Manga Wildlife Park
- Clifton Fish Aquarium
- Hyderabad Zoo
- Islamabad Zoo
- Jallo Wildlife Park

- Jungle World
- Karachi Municipal Aquarium
- Karachi Safari Park
- Karachi Zoo
- Lahore Walkthrough Aviary
- Lahore Zoo
- Lahore Zoo Safari
- Landhi Korangi Aquarium
- Landhi Korangi Zoo

114) Collapse disorder?

Bees are responsible for pollinating more than 250,000 species of plants. "Colony collapse disorder" is wiping out entire populations of the insect. Scientists have yet to discover its true cause. Continued losses could threaten the supply of crops like almonds, apples and cucumbers. From all species, humans rely upon for food 87 rely on pollinators mainly honeybees. Only 28 different crops could survive without such assistance.

115) Protected area?

"A clearly defined geographical space, recognized, dedicated and managed, through legal or other effective means, to achieve the long-term conservation of nature with associated ecosystem services and cultural values".

116) Pakistan Terrestrial water and maritime zone act 1976?

Pakistan Terrestrial Water & Maritime Zones Act, 1976;

- It includes provisions for preservation, development and protection of marine environment
- This law controls marine pollution and exploration, development, conservation and management of living resources in Pakistan's Exclusive Economic Zone (EEZ)
- This law means that a ship carrying nuclear and hazardous substances will have to inform the Government of Pakistan.

117) Primary objective of National Park?

Primary objective

To protect natural biodiversity along with its underlying ecological structure.

118) Game species?

Most of the areas in game reserves have created to provide habitat protection for animal species commonly referred to as game species (huntable species for sport or meat)

119) Conservation status of Hingol National park?

Hingol National Park

- Hingol National Park spread over an area of about 1,650 square km along the Makran Coast, Baluchistan
- It is the largest of National Parks of Pakistan
- The area was for the first time declared reserved in 1988.
- Hingol is known to support threatened invertebrates in addition to a variety of bird species
- The park is an excellent habitat to wild animals including over 3000 ibexes, and 1500 Urals and more than 1200 Chinkara
- A number of resident and migratory birds are supported by this park.

120) Status of World's Forest Genetic Resources?

In 2014, the Food and Agriculture Organization of the United Nations published the first State of the World's Forest Genetic Resources. The publication addressed the conservation, management and sustainable use of forest tree and other woody plant genetic resources of actual and potential value for human well-being in the broad range of management systems.

121) Freezing methods of cryopreservation?

Freezing

Cryopreservation is based on the ability of certain small molecules to enter cells and prevent dehydration and formation of intracellular ice crystals, which can cause cell death and destruction of cell organelles during the freezing process. The sensitivity of cells to low temperature depends on the plant species. There are four different

Types of methods :

1. **Slow freezing method** - the tissue or plant material is slowly frozen at slow cooling rate. The advantage is the plant cells are partially dehydrated and survive better.
2. **Rapid freezing method** - it involves plunging the vials in liquid nitrogen. The temperature decreases from -300 to -1000 degree rapidly.
3. **Combined freezing method** - this is combination of both slow and rapid freezing method. The process is carried out in step wise like manner.
4. **Dry freezing method** - in this method dehydrated cells and seeds are stored.

122) Role of adaptation and standing genetic variation in response of trees population to environmental changes?

Adaptation and standing genetic variation

Genetic adaptations that make a population more suited for survival are achieved through gene frequency changes across generation. Many tree species have high genetic variability in adaptive traits and can therefore grow under a wide range of conditions. Indeed, phenotypic traits of adaptive importance, such as drought tolerance, cold-hardiness, resistance to pests and diseases, and flowering and fruiting period, have been shown to vary across ecological and geographic gradients to an extent that may be as important as the differences observed amongst species.

123) One example of bottle neck effect?

Northern Elephant seal Example:

Northern elephant seals have reduced genetic variation most likely due to being hunted. Population bottleneck in 1890s due to overhunting. Hunting reduced their population size to as few as 20 individuals at the end of the 19th century. Now over 30,000, very little variation left in this population.

124) Can microorganisms can face harsh conditions?

Microbes cling to life in some of Earth's most extreme environments, from toxic hot springs to high-altitude deserts. These 'extremophiles' include organisms that can survive near-boiling heat or near-freezing cold, high pressure or high salt, as well as environments steeped in acids, alkalis, metals or radioactivity

125) Difference between policy and strategy?

A **Strategy** is a special plan made to achieve a market position and to reach the organizational goals and objectives, but **Policy** refers to a set of rules made by the organization for rational decision making. Many people have confusion regarding the two terms, but they are not alike.

126) Name sanctuary of Pakistan?

Wildlife Sanctuaries of Pakistan are;

- Astor Wildlife Sanctuary
- Baltistan Wildlife Sanctuary
- Chasma and Taunsa Barrage Dolphin Sanctuary
- Cholistan Wildlife Sanctuary
- Hab Dam Wildlife Sanctuary
- Kargah Wildlife Sanctuary

127) Features of national monument?

Distinguishing features

Category III protected areas are usually relatively small sites that focus on one or more prominent natural features and the associated ecology, rather than on a broader ecosystem.

128) PGRFA?

Plant genetic resources

Plant Genetic Resources for Food and Agriculture (PGRFA) are the raw material that farmers and plant breeders use to improve the quality and productivity of crops. They can be defined as any genetic material of plant origin of actual or potential value for food and agriculture, e.g. seeds, tubers, mature plants etc.

129) Worlds forest conservation?

- These resources are also the basis for large-scale wood production in planted forests to satisfy the worldwide need for timber and paper.
- Genetic refers to variation of genetic DNA origin, and variation of genes at different levels:
 1. Variation between species,
 2. Variation between populations within species
 3. Variation between individual trees within populations. The largest variation is between species, and loss of whole species is therefore also the most dramatic loss of future options.
- Resources refers to the use of genetic variation—in the broad sense stated above—considered to be of potential value for humans at present or in the future.

The State of the World's Forest Genetic Resources

In 2014, the Food and Agriculture Organization of the United Nations published the first State of the World's Forest Genetic Resources .The publication addressed the conservation, management and sustainable use of forest tree and other woody plant genetic resources of actual and potential value for human well-being in the broad range of management systems.

130) Role of botanical gardens in conservation?

Conservation

Conservation of rare and threatened plants. The conservation of plant diversity is critical for sustainable development and botanic gardens are playing a key role as centres of conservation action. Botanical gardens can promote diversity. Because they include many species of plant.

131) Why Indus dolphins are becoming extinct day by day?

The Indus River Dolphin

The Indus River dolphin is one of the world's rarest mammals. It is second most endangered freshwater river dolphin. Approximately 1,100 specimens of this species exist today in a small fraction of their former range. Population of this species has gradually declined due to various

factors e.g. water pollution, poaching, fragmentation of habitat due to barrages and dolphin stranding in the irrigation canals.

132) Demerits of ex situ and In-situ?

In situ Conservation

Merits:-

- Plants and animals conserved in their natural environment
- Biodiversity permanently protected
- Natural and cultural heritage protected permanently
- Ecological integrity is maintained and managed
- Opportunities may arise for ecologically sustainable land uses (which come with associated economic benefits)
- Facilitates scientific research of the site

Demerits:-

- Genetic diversity may have already been dramatically decreased
- Conditions that threatened the organisms in the area may still be present, e.g. disease or interspecific competition
- Poachers and Eco tourists may see the thriving area as an opportunity and may cause damage

Ex situ Conservation

Merits:-

- It can be used to protect individual animals in a controlled environment. This means that issues such as predation and hunting can be monitored and managed more easily.
- It can be used to reintroduce species that have left an area.

Demerits:-

- Usually only a small number of individuals can be cared for.
- It can be difficult and expensive to create and sustain the right environment.
- The animals that are habituated (used to) human contact may be less likely to exhibit natural behaviours and may be more likely to catch a disease from humans.
- This type of conservation is usually less successful as many species can't breed successfully in captivity or don't adapt to their new environment when moved to a new location.

133) What is main national program for PGR?

The Plant Genetic Resources Program (PGRP) is the main National Program that is functioning for overall management of PGRs including exploration, collection, exchange, safe storage on long, medium and short term basis, evaluation, documentation and distribution to other research institutes for sustainable utilization of plant genetic resources of crops and wild relatives within the country and abroad. Other research and educational institutes both at Federal and Provincial level are also handling conservation and use of plant genetic resources of selected crop commodities or groups of plant species. Attack oil refinery in Rawalpindi has also developed a biodiversity park as an environment improvement strategy for compensating the pressure on biodiversity due to industrial processing in the nearby vicinity. Institute of Agricultural Biotechnology and Genetic Resources (IABGR). PGRP is regularly funded by the government of Pakistan.

Plant Genetic Resources Institute hosts the sole National Genebank of Pakistan for conservation of plant genetic resources and six labs including

1. Germplasm exploration lab
2. Seed preservation lab
3. In vitro conservation lab
4. Germplasm evaluation lab
5. Plant introduction and seed health lab
6. Data management lab

134) What is the criteria of national park?

Distinguishing Features

The area should contain representative examples of major natural regions, and biological and environmental features or scenery. It should be of sufficient size to maintain ecological processes. The composition, structure and function of biodiversity should be to a great degree in a “natural” state.

135) Mission of Ramsar convention?

Mission

The Convention’s mission is

“The conservation and wise use of all wetlands through local and national actions and international cooperation, as a contribution towards achieving sustainable development throughout the world.

136) Link between forest genetic resources (FGR) and climate change?

Plants can alter the temperature of the Earth’s atmosphere. Through the process of photosynthesis, plants use energy from the sun to draw down carbon dioxide from the

atmosphere and then use it to create the carbohydrates they need to grow. Since carbon dioxide is one of the most abundant greenhouse gases, the removal of the gas from the atmosphere may temper the warming of our planet as a whole. Transpiration in plants can increase water vapour in the atmosphere, causing more precipitation and cloud cover in an area. The additional cloud cover often reinforces the cooling by blocking sunlight. Contribute to soil fertility and prevent soil erosion.

137) Habitat of Species That Affect Climate Change?

Global warming resulting from human emissions of greenhouse gases. The consequences include habitat loss; shifts in climatic conditions and in habitats that surpass migrational capabilities; altered competitive relationships.

138) How Plant Cope Changes Climate?

Maintaining insect species that can provide pollination services for a wide range of crops is also vital to the future of agriculture in the face of climate change. Pollinator populations not only need to be able to cope with changing climatic conditions, they must also be able to provide the pollination services needed to meet increasing demands for food and retain the capacity to adapt to potential changes in the types of crops grown.

139) Impact of Climate Changes On FGR? And Climate Effect Organisms Which Are Changes Associate with Tree?

Climate change may also result in high variability in temperature and precipitation, with an increase in incidence of extreme events, such as flooding, late frosts and intensive summer droughts, amongst other events. In some areas, such as the Mediterranean and the Neo-tropics, an increase in seasonality is also expected. Under such conditions, natural selection may not result in efficient adaptation because selection pressures are multi-directional, involving traits that may be inversely correlated at the gene level. The standing genetic variation in populations may then not be large enough to create the rare new genotypic combinations that are required.

Ecosystems affected by abrupt change may sustain rapid and widespread transformation as ecological tipping points are exceeded. Given the pivotal role of trees in ecosystem function, abrupt climate change impacts on them may thus have profound consequences for forests as a whole. Irreversible loss of ecosystem integrity and function may follow, with replacement by new no endemic ecosystems.

Effects of changing climate on organisms associated with trees in particular, changes in the biology of insect pests and diseases may make ecosystems more susceptible to tree mortality. Because of improved environmental conditions for the pest and reduced tree resistance due to increased stress, pests may react to climate change with range expansions and/or increases in attack severity.

140) What Is the Extreme Weather Effect on Invertebrate's Genetic Resources?

Because of the many ecosystem services that they provide, invertebrates have a key role to play in adapting agriculture to the effects of climate change. The extent to which the individual services provided by invertebrates will be enhanced or impeded by climate change is difficult to predict. However, if invertebrate biodiversity is lost, the capacity of ecosystems to adapt is likely to diminish.

Healthy soils – and healthy, diverse soil invertebrate communities – will be vital to climate change adaptation. For example, Earthworms help to maintain soil structure and the availability of water throughout the soil profile. Studies have shown that the presence of these animals can help to alleviate the effects of drought on crop Production. Studies have also revealed the remarkable. Ability of diverse soil invertebrate communities to restore the structure of degraded soil. He potential for managing soil invertebrates to enhance their beneficial roles has been little explored. Few if any deliberate attempts have been made to introduce soil invertebrates into new countries or ecosystems. Given the potential for such species to become invasive, it is inadvisable to attempt any such introductions until soil ecology is much better understood than it is today. However, every effort should be made to avoid agricultural practices that disrupt resident soil invertebrate communities and the services they provide.

It is likely that some pests, as they move into new areas in response to climate change, will at least temporarily escape from their natural enemies. This is likely to increase demand for classical biological control agents, especially in places where the newly established pest population is separated from its original home by a physical barrier such as the sea or a mountain range. For this reason, access to new classical biological control agents is likely to be particularly important for island countries

141) Discuss the Impact of Climate Change On FGR. Also Discuss How This changing Climate Effect Organism Which Is Associated with Trees?

Forest genetic resources are essential for forest-depending communities who rely for a substantial part of their livelihoods on timber and non-timber forest products (for example fruits, gums and resins) for food security, domestic use and income generation.

Forest Genetic Resources and Climate Change

Diversity of forest genetic resources enables the potential for a species (or a population) to adapt to climatic changes and related future challenges such as temperature changes, drought, pests, diseases and forest fires. Though forest trees are known for showing great plasticity in their response to climate changes, not all species are naturally capable to adapt at the pace necessary.

Climate change may also result in high variability in temperature and precipitation, with an increase in incidence of extreme events, such as flooding, late frosts and intensive summer droughts, amongst other events. In some areas, such as the Mediterranean and the Neo-tropics, an increase in seasonality is also expected. Under such conditions, natural selection may not result in efficient adaptation because selection pressures are multi-directional, involving traits that may be inversely correlated at the gene level. The standing genetic variation in populations may then not be large enough to create the rare new genotypic combinations that are required.

Ecosystems affected by abrupt change may sustain rapid and widespread transformation as ecological tipping points are exceeded.

Given the pivotal role of trees in ecosystem function, abrupt climate change impacts on them may thus have profound consequences for forests as a whole. Irreversible loss of ecosystem integrity and function may follow, with replacement by new non endemic ecosystems.

Direct Impacts of Climate Change

These include high tree mortality through extreme climatic events, particularly drought in combination with widespread regeneration failure, for example, examined the evidence for anthropogenic climate change leading to future large-scale “dieback” in Amazonian rain forest. Analysis suggested that dry season water stress is likely to increase in eastern Amazonia over the 21st century, with the region tending toward a climate more appropriate to seasonal forests.

Effects of Changing Climate on Organisms Associated with Trees In particular, changes in the biology of insect pests and diseases may make ecosystems more susceptible to tree mortality. Because of improved environmental conditions for the pest and reduced tree resistance due to increased stress, pests may react to climate change with range expansions and/or increases in attack severity.

Changes in Abiotic Disturbance Regimes:

These include changes in fire regimes, flooding, landslides and/or hurricanes. Fire and climate are closely linked and are also associated with changes in land use. Coupled climate and fire-risk models suggest not only an increase in the frequency of fires but also in fire size and length of the fire-risk season, with some areas subject to risk that were not before. Malhi et al. (2009) considered how tipping points may be reached in Amazonian rainforest by a combination of increased dryness and an increased incidence of fire events

Invasion by organisms foreign to local ecosystems

This includes the increased risk of establishment by invasive species which accidentally arrive into ports of entry, through globalized commerce. By making new niches available, climate change will facilitate the survival of mammals, insects, diseases and/or weeds foreign to endemic ecosystems.

142) Background history of Ramsar conservation?

History:

Ramsar is one of the global inter-governmental environmental agreements. The treaty was negotiated in 1960s by countries and NGOs. To avoid the increasing loss and degradation of wetland habitat for migratory water birds. In an 18 nations meeting it was adopted in the Iranian city of Ramsar On 2nd February 1971. Came into force in 21st December 1975

Past finals – 2021, 2020, 2019, 2018

143) What is plant genetic resources conservation or germplasm conservation?

Germplasm or plant genetic resource has to be maintained in such a state that minimize the risk of loss and allows either is direct planting in the field or its preparation of planting with relative ease this is called plant genetic resources conservation or germplasm conservation

144) Why do we need to conserve plant genetic resources?

- Conservation of plant genetic resources is necessary for food security and agro-biodiversity
- Biodiversity provides a valuable source of compounds to the medical, food and crop protection industries.
- Maintenance of ecosystem
- Genetic resources need to be conserved so that they may be used in crop research and be used as sources of genes for crop improvement

145) What is Conservation of plant genetic resources?

- Selection of target taxa
- Project commission
- Eco geographic survey/preliminary survey mission
- Conservation objectives
- Field exploration
- Conservation strategies
- Conserved product deposition and dissemination
- Characterization/ Evaluation
- PGR utilization
- Utilization products

146) Game Reserve?

“A game reserve is an area wherein controlled hunting and shooting is permitted on permit basis”. A game reserve (wildlife preserve) is a large area of land where wild animals live safely or are hunted in a controlled way for sport. In the game reserves the major focus is specifically the animals.

147) Name of second largest national park?

Kirthar National Park is the second largest national park of Pakistan spread over an area of 3000 square kilometres.

148) Define strict nature reserve?

Protected areas that are strictly set aside to protect biodiversity where human visitation, use and impacts are strictly controlled to ensure protection of the conservation values.

149) Define phenotypic plasticity?

Phenotypic plasticity is defined as the capacity of a particular genotype to express different phenotypes under different environmental conditions.

150) Assisted migration?

Assisted migration involves human movement of tree seed and seedlings from current locations to sites modelled to experience analogous environmental conditions in the future. Assisted migration may be undertaken over long distances, or just beyond the current range limit of particular genotypes and populations, or within the existing range.

151) Central goal of zoo?

Unfortunately not all zoos are scientific in nature, and extreme controversy has arisen regarding how the animals are treated. Suffice to say, regulation is necessary to ensure proper care. Conservation (not exploitation) should always be the central goal behind any legitimate zoo. Zoos provide the opportunity for people to see a glimpse of this side of nature. Zoo plays important role in conservation of many threatened/endangered species.

152) Mission of Ramsar Convention?

The Convention's mission is "The conservation and wise use of all wetlands through local and national actions and international cooperation, as a contribution towards achieving sustainable development throughout the world."

153) Role of earthworm in soil?

Earthworms help to maintain soil structure and the availability of water throughout the soil profile. Studies have shown that the presence of these animals can help to alleviate the effects of drought on crop. Production. Studies have also revealed the remarkable ability of diverse soil invertebrate communities to restore the structure of degraded soil. The potential for managing soil invertebrates to enhance their beneficial roles has been little explored.

154) What is legislation?

The process of making or enacting laws.

155) Types of gene bank?

Under suitable conditions genetic resources are conserved for a long term as gene bank. Such gene bank is of two types:

1. In vivo Gene Bank
2. In vitro Gene Bank

156) Disadvantages of DNA bank?

There are problems with subsequent gene isolation, cloning and transfer of DNA back to a plant and it currently does not allow the regeneration of the same genotype as the original sample.

Clonal repository?

Clonal repository is field Gene bank where genetic resources of clonally propagated crops like fruits are preserved as living plants. Various institutions are involved in the capacity building to develop AnGR, in the country.

157) Two types of cryoprotectants?

Membrane permitting which can freely diffuse the membrane such as glycerol (G), ethylene glycol (EG) and dimethyl sulfoxide (DMSO)

Non Membrane permitting which cannot permeate the cell membrane such as sugars

158) Maximum temp 4 cryopreservation?

Maximum temperature of cryopreservation is -196C.

159) Different variation of gene variation?

Genetic refers to variation of genetic DNA origin, and variation of genes at different levels:

1. Variation between species,
2. Variation between populations within species
3. Variation between individual trees within populations. The largest variations between species, and loss of whole species is therefore also the most dramatic loss of future options.

Resources refers to the use of genetic variation—in the broad sense stated above—considered to be of potential value for humans at present or in the future.

160) Name of physical growth limitations in median term storage?

Physical growth limitation

- Low temperature
- Low light/restricted photoperiod
- Minimal containment
- Minimal O₂
- Osmotic (water) stress

161) Objective of category 6?

Primary objective

- To protect natural ecosystems.
- Use natural resources sustainably, when conservation and sustainable use can be mutually beneficial.

162) Biodiversity of habitat- species management area?

Definition: IUCN Management Category IV (Habitat/Species Management Area) refers to areas that are managed to protect particular species or habitats. They are defined by IUCN as “protected areas aiming to protect particular species or habitats and management reflect this priority.

163) State of worlds forest genetic resources?

In 2014, the Food and Agriculture Organization of the United Nations published the first State of the World's Forest Genetic Resources .The publication addressed the conservation, management and sustainable use of forest tree.

164) Ramsar sites of Panjab?

- Uchhali Complex
- Taunsa Barrage
- Chashma Barrage

165) Criteria of national park?

Distinguishing features

- The area should contain representative examples of major natural regions, and biological and environmental features or scenery.
- It should be of sufficient size to maintain ecological processes.
- The composition, structure and function of biodiversity should be to a great degree in a “natural” state.

166) Who is responsible for botanical garden?

Botanical gardens are often run by universities or other scientific research organizations, and often have associated herbaria and research programmers in plant taxonomy or some other aspect of botanical science.

167) Role of national park in landscape and seascape?

- Protecting some of the earth's richness that will not survive outside.
- Protecting additional ecosystem services.
- Providing areas where ecosystems can be studied in as pristine an environment as possible.
- Protecting natural sites that are also of religious and cultural significance.

168) CBD history?

Cartagena protocol was adopted on June 2001 in Cartagena, Spain. It entered into force on September 11th; 2003. Pakistan signed the Cartagena protocol in June 200. Pakistan has ratified it in May 2009.

169) Three pillars of Ramsar site?

The “three pillars” of the convention

The Contracting Parties (160) commit to:

- Work towards the wise use of all their wetlands.
- Designate suitable wetlands for the list of Wetlands of International Importance.
- Cooperate internationally on transboundary wetlands, shared wetland systems and shared species.

170) Founder effect 3 examples?

In population genetics, the founder effect is the loss of genetic variation that occurs when a new population is established by a very small number of individuals from a larger population

The Amish People

Around 200 German immigrants settled in Pennsylvania within community marriages. Developed syndrome named Ellis-van Creveld syndrome.

Sickle Cell Disease

For most of humanity's existence, sickle cell disease usually meant an early death, most likely as a young child. (It still does in underdeveloped nations.) In fact, the average life span for a sufferer in the US in 1973 was only 14 years. Now it's 40–60 years in the US. The cause of this disorder: genetic changes meant to protect against malaria. As a result, those who suffer from sickle cell disease overwhelmingly come from tropical areas or places where malaria is common.

Huntington's disease

Huntington's disease (aka Huntington's chorea) is a genetic disorder which results in slowly progressing brain cell death. There are two distinct populations in which the disorder occurs much more often.

1. The first group is the Afrikaner population of South Africa.
2. The second group is the residents of the Lake Maracaibo region of Venezuela.

171) Three causes of extinction?

Climate Change

Almost half of plant and animal species have experienced local extinctions due to climate change. Global warming could trigger not just local but global extinctions of animals and plants. Species already threatened by habitat destruction, pollution, alien invasion and overhunting are

more vulnerable to climate change. Diversity of species in any one ecosystem could be affected by rises in average temperatures or a shift of climate regime

Habitat destruction

Deforestation has killed off more species than we can count. Rainforest can disappear in the next 100 years if deforestation is not stopped. 13 million hectares of forest have been converted or destroyed. Coral reefs are also threatened. Reefs are home to 25% of marine animal. To date, 27% of coral reefs have been destroyed.

Lack of genetic diversity

When species starts dwindling in numbers, there's a smaller pool of available mates. Dwindling population of African cheetah's suffers from unusually low genetic diversity. Thus may lack the resiliency to survive another major environmental disruption.

172) Storage Strategy of DNA banks?

Storage strategy

Determining what to store and for how long is an important consideration, used to determine sample size, capacity of the DNA bank, preparation of samples and documentation. Long-term needs and expected volume and number of samples to be stored will determine organization and repository design.

173) Objectives of CBD?

Objectives of CBD, 1992

The three inter-related objectives are:

- The conservation of biological diversity;
- The sustainable use of its components;
- The fair and equitable sharing of the benefits arising out of the utilization of genetic resources, including by appropriate
- Access to genetic resources,
- Transfer of relevant technologies,
- Funding.

174) Three botanical gardens?

1. Cacti and succulent plants.
2. Herb gardens.
3. Greenhouses, shade houses.

175) Application of cryopreservation?

In Animal Husbandry

The introduction of cryopreservation technology leads a major breakthrough in animal husbandry. Since the 1st successful cryopreservation of bull semen has been used to propagate the rare and endangered species using assisted reproduction techniques.

In fishery science

The 1st report on fish sperm cryopreservation was published by Blaxter (1953). To date milt (semen) of over 200 species of fresh water and marine fish have been cryopreserved and have been adequate for the purpose of cryo banking (10, 11, 12). In fish aquaculture the successful cryopreservation of gametes and embryos could offer new commercial possibilities, allowing the unlimited production of fry and potentially healthier and better conditioned fish as required

In medical science

Low temperature have been used in medicine and to prevent food spoilage since ancient time. Now- a- days it is used in fertility treatment the transport of human organs and the long- term storage of biological specimens, either for future or simply as a record of biodiversity.

Cryopreservation of testicular tissue

Cryopreservation of immature testicular tissue is a developing method to avail reproduction to young boys who need to have gonado toxic therapy

Embryo cryopreservation

Embryo cryopreservation is used most often to store good quality excess embryos resulting from an IVF treatment cycle. Embryos can be stored for a patient who elects to have her eggs fertilized with donor sperms. Pregnancies have been reported from embryos stored for 16 years

176) Processing of sample in DNA bank?

DNA preserved in DNA banks will be stored either within cells and extracted upon retrieval from storage or extracted from cells and purified before storage. The quality of the DNA is expressed through yield, purity, molecular weight, amplification efficiency and authenticity of sequences. The quality of DNA extracted from plant specimens is dependent on the condition of the specimen before storage, the storage environment and the duration of storage. Rapid drying of plant samples with silica gel or lyophilisation helps to preserve the DNA.

177) Features of habitat species management area?

- Protection of particular species
- Protection of habitats
- Active management to maintain target species
- Active management of culturally-defined ecosystems

178) Link between climate change and botanical garden?

Plants can alter the temperature of the Earth's atmosphere. Through the process of photosynthesis, plants use energy from the sun to draw down carbon dioxide from the

atmosphere and then use it to create the carbohydrates they need to grow. Since carbon dioxide is one of the most abundant greenhouse gases, the removal of the gas from the atmosphere may temper the warming of our planet as a whole. *transpiration in plants can increase water vapour in the atmosphere, causing more precipitation and cloud cover in an area. The additional cloud cover often reinforces the cooling by blocking sunlight. *Contribute to soil fertility and prevent soil erosion.

179) Pakistan terrestrial water and marine zone act?

It includes provisions for preservation, development and protection of marine environment

- This law controls marine pollution and exploration, development, conservation and management of living resources in Pakistan's Exclusive Economic Zone (EEZ)
- This law means that a ship carrying nuclear and hazardous substances will have to inform the Government of Pakistan.

180) PEPA provides for;

- Protection
- Conservation,
- Rehabilitation
- Improvement of the environment;
- PEPA provide framework for prevention and control of pollution
- Helps in protection of sustainable development.

181) Effect of climate change on invertebrate genetic resource?

Extreme weather events such as heat waves, droughts and floods - which are predicted to increase in frequency due to climate change – are often followed by pest outbreaks. Among other contributing factors, these outbreaks can occur because the extreme event eliminates or weakens pest's natural enemies.

182) Purpose of CBD?

- CBD is about the conservation and wise use of different biological resources (plants and animals).
- It was adopted in 1992 at Rio De Janeiro, Brazil and entered force on January, 1993, which was 90 days after the 30th ratification.
- Pakistan signed it in June 1992 at United Nations Conference on Environment and Development held at Rio De Janeiro, Brazil
- Pakistan ratified it on 26th July 1994.
- The Convention on Biological Diversity covers biodiversity at all levels:
- Ecosystems

- Species
- Genetic resources
- It also covers biotechnology, including through the Cartagena Protocol on Biosafety.
- In fact, it covers all possible domains that are directly or indirectly related to biodiversity and its role in development, ranging from science, politics and education to agriculture, business, culture and much more.

183) Strategies followed by effective conservation of AnGR?

Following strategies should be followed for effective conservation and utilization of AnGR:

1. Formulating the National Livestock and wild-life Breeding Policies.
2. Encouraging the Formation of Breed Associations.
3. Developing Professional Human Resources.
4. Strengthening Research and Development Institutions
5. Developing Infrastructure for marketing International co-operation and assistance is needed in capacity building to remodel available livestock farms/research stations to conserve and develop genetic resources

184) Future of cryopreservation?

Vitrification method of cryopreservation may bring new opportunities to research protocols. It is still an experimental procedure. There are two major concern about vitrification - toxicity of high concentration of cryoprotectants used and microbial contamination of liquid nitrogen. Several IVF programs have adopted the vitrification method as the sole procedure for day-3 human embryos and for human blastocysts, with excellent survival and pregnancy rates. The challenge now is to find a protocol to successfully vitrify human oocytes for which the slow freezing method has yet to produce acceptable.

185) Why do forest genetic resources matter?

The high levels of genetic variation that are present within many tree species can be beneficially developed and used by foresters and tree growers. Whereas agricultural crop breeders and farmers often substantially modify the growing environment to suit a specific crop species or variety, tree growers commonly identify species and provenances which can provide some improved levels of the goods and services required even without intensive selection and improvement, or intense management requirements, or major modification of the external environment.

186) Cryopreservation method?

There are four different types of methods:

- 1 Slow freezing method- the tissue or plant material is slowly frozen at slow cooling rate. The advantage is the plant cells are partially dehydrated and survive better.

2. Rapid freezing method - it involves plunging the vials in liquid nitrogen. The temperature decreases from -300 to -1000 degree rapidly.
3. Combined freezing method - this is combination of both slow and rapid freezing method. The process is carried out in step wise like manner.
4. Dry freezing method - in this method dehydrated cells and seeds are stored.

187) Animals included in quarantine?

Animal and plant quarantine programs are intended to protect agriculture from the threat of entry of exotic hazardous organisms. In some countries this objective may be extended to the protection of natural domestic flora and fauna. Both types of programs regulate the importation of living individuals. The general concepts and objectives of plant and animal quarantine are similar; but differences in biology, agricultural production, marketing, exporting, and importing necessitate a variety of quarantine procedures. Animal and plant quarantine procedures. Animal and plant quarantine programs are intended to protect agriculture from the threat of entry of exotic hazardous organisms. In some countries this objective may be extended to the protection of natural domestic flora and fauna.

188) Function of invertebrates?

Because of the many ecosystem services that they provide, invertebrates have a key role to play in adapting agriculture to the effects of climate change. The extent to which the individual services provided by invertebrates will be enhanced or impeded by climate change is difficult to predict. However, if invertebrate biodiversity is lost, the capacity of ecosystems to adapt is likely to diminish.

Healthy soils – and healthy, diverse soil invertebrate communities – will be vital to climate change adaptation. For example, earthworms help to maintain soil structure and the availability of water throughout the soil profile. Studies have shown that the presence of these animals can help to alleviate the effects of drought on crop Production. Studies have also revealed the remarkable ability of diverse soil invertebrate communities to restore the structure of degraded soil. The potential for managing soil invertebrates to enhance their beneficial roles has been little explored. Few if any deliberate attempts have been made to introduce soil invertebrates into new countries or ecosystems. Given the potential for such species to become invasive, it is inadvisable to attempt any such introductions until soil ecology is much better understood than it is today. However, every effort should be made to avoid agricultural practices that disrupt resident soil invertebrate communities and the services they provide.

In the case of classical biological control agents, the genetic diversity of introduced populations may be relatively low because the introduction was based on a small founder population. This lack of diversity may inhibit the ability of the population to respond to climate change. It is likely that some pests, as they move into new areas in response to climate change, will at least temporarily escape from their natural enemies. This is likely to increase demand for classical biological control agents, especially in places where the newly established pest population is

separated from its original home by a physical barrier such as the sea or a mountain range. For this reason, access to new classical biological control agents is likely to be particularly important for island countries.

189) Laws of environmental resources?

Some major laws dealing with different resources present in environment are as follow;

- Pakistan Environmental Protection Act, 1997
- Pakistan Penal Code, 1869• Forest Act, 1927
- Pakistan Terrestrial Water & Maritime Zones Act, 1976

1. Pakistan Environmental Protection Act, 1997

PEPA provides for;

- Protection,
- Conservation,
- Rehabilitation and
- Improvement of the environment;
- PEPA provide framework for prevention and control of pollution
- Helps in protection of sustainable development.

2. Pakistan Panel Code, 1860

The polluter of the environment can be punished under this code for certain types of pollution.

These punishment are of following types;

- Punishment for water pollution
- Punishment for atmospheric pollution
- Punishments for general pollution

3. Forest Act, 1927

- This law empowers provincial governments to manage forests under their area.
- The government can reserve the state-owned forest land, assume control of privately owned forest land and declare any government owned land in a protected area.
- It prohibits the clearing of forest for cultivation, grazing, hunting, removing forest produce, quarrying and felling.

4. Pakistan Terrestrial Water & Maritime Zones Act, 1976;

- It includes provisions for preservation, development and protection of marine environment
- This law controls marine pollution and exploration, development, conservation and management of living resources in Pakistan's Exclusive Economic Zone (EEZ)
- This law means that a ship carrying nuclear and hazardous substances will have to inform the Government of Pakistan.

190) Quarantine principle?

One recent study (Plucknett and Smith, 1988) describes six principles of successful quarantine. They are summarized as follows

1. **Sound scientific and technical principles** should form the foundation of quarantine program. Pests and pathogens should be ranked by quarantine services according to the potential danger they pose to crops and the potential for success in excluding them. For example, germplasm from centres of diversity should receive a high priority because of the potential for such accessions to harbour coevolved pests or pathogens.

2. **Animal and plant quarantine regulations** are similar in that they may:

Require import permits issued by the quarantine service of the importing country (these may require the exporting country to certify that specified conditions have been met prior to shipment);

- Specify things that are prohibited from entry;
- Grant exceptions to the prohibitions for scientific purposes;
- Require inspection of imported materials upon arrival;
- Require appropriate treatment, if warranted, as a condition of entry; and
- Require, after arrival, quarantine or isolation in an approved facility.

3. **When germplasm must be planted** and grown for the purposes of quarantine testing, it should be done in an area geographically and ecologically separated from the major growing areas for that crop, to prevent the establishment of crop-specific pests or pathogens.

4. **When germplasm is endangered** or the need for particular accessions is particularly urgent, some discretion should be possible on the part of quarantine officials in allowing exceptions for controlled entry, despite existing regulations to the contrary.

5. **Decentralized quarantine services** are generally more efficient because they enfold a wider range of expertise in germplasm assessment.

6. **Because delays in transit** can be detrimental for any germplasm accessions, access to good communication and transportation services is essential for quarantine.

191) Hingol national park?

Hingol National Park spread over an area of about 1,650 square km along the Makran Coast, Balochistan. It is the largest of National Parks of Pakistan. The area was for the first time declared reserved in 1988. Hingol is known to support threatened invertebrates in addition to a variety of bird species. The park is an excellent habitat to wild animals including over 3000 ibexes, and 1500 Urinals and more than 1200 Chinkara. A number of resident and migratory birds are supported by this park.

192) Write the impact of climate change on FGR also discuss the effect of climate change on organism?

Climate change may also result in high variability in temperature and precipitation, with an increase in incidence of extreme events, such as flooding, late frosts and intensive summer droughts, amongst other events. In some areas, such as the Mediterranean and the Neo-tropics, an increase in seasonality is also expected. Under such conditions, natural selection may not result in efficient adaptation because selection pressures are multi-directional, involving traits that may be inversely correlated at the gene level. The standing genetic variation in populations may then not be large enough to create the rare new genotypic combinations that are required. Ecosystems affected by abrupt change may sustain rapid and widespread transformation as ecological tipping points are exceeded. Given the pivotal role of trees in ecosystem function, abrupt climate change impacts on them may thus have profound consequences for forests as a whole. Irreversible loss of ecosystem integrity and function may follow, with replacement by new non endemic ecosystems.

Effects of changing climate on organisms associated with trees

In particular, changes in the biology of insect pests and diseases may make ecosystems more susceptible to tree mortality. Because of improved environmental conditions for the pest and reduced tree resistance due to increased stress, pests may react to climate change with range expansions and/or increases in attack severity.

193) Name of second largest park?

Kirthar National Park is the second largest national park of Pakistan spread over an area of 3000 square kilometres. Kirthar was designated a national park by the Sindh Wildlife Department in 1974, This is the first of Pakistan's parks to be included in the UN's listing of National Parks of 1975 This is natural haven for Urial sheep, Ibex, Chinkara gazelle, Jungle cats, desert cats, occasional leopard, desert wolf also prowl the park.

194) What is a National Strategy for PGRFA?

A National Strategy for PGRFA is the blueprint for the management of a country's PGRFA as a continuum of interventions in order to achieve clearly defined time bound goals. A well-designed National Strategy needs to be tailored to the particular

195) Why we need to preserve Aquatic Genetic resources?

If human fail to preserve aquatic genetic resources, it will cost in diminished quality of life” protein for human consumption worldwide. Fishing industry plays a major role in socioeconomic terms. Developing countries local fishing and the related branches of production guarantee the income of a large part of coastal communities.

196) Transfer of DNA in bacteria horizontally?

There are three ways for bacteria to transfer their DNA horizontally

Conjugation: The transfer of DNA directly from one cell to another through cell- cell contact often involving plasmids

Transformation: Bacteria are capable of taking up DNA directly from their environment and incorporating it into their genomes known as natural transformation

Transduction

Transduction is the transfer of DNA from one cell to another by a virus.

197) Vertical gene Transfer?

The transfer of genes from parents to offspring. In vertical gene transfer, the transfer of genetic material is from parents to offspring. It may be through sexual or asexual reproduction. Acquisition of DNA through horizontal gene transfer is distinguished from the transmission of genetic material from parents to offspring during reproduction, which is known as vertical gene transfer.

198) Grading up?

Breeding of animals of two different breeds where the animals of an indigenous breed/genetic group are mated by an improved pure breed for several generations towards attaining the superior traits of the improved breed. Grading up is continuous use of purebred sires of the same breed in a grade herd. By fifth generation, the graded animals may reach almost purebred levels.

199) Allopatric speciation?

“Gene flow blocked by physical barriers results in Allopatric speciation” It is geographical isolation that doesn’t allow population of the same species to exchange genetic material . Physical barriers to gene flow both “natural” and “artificial

200) Inbreeding depression?

Inbreeding depression is the reduced biological fitness in a given population as a result of inbreeding, or breeding of related individuals. In a small population, mating between relatives are common. This inbreeding may lower the population’s ability to survive and reproduce, a phenomenon called inbreeding depression.

201) Biometric gene environment interaction?

The biometric (or statistical) conception has its origins in research programs that seek to measure the relative proportions of genetic and environmental contributions to phenotypic variation within populations. Biometric interaction is relevant in the context of research on individual differences rather than in the context of the development of a particular organism.

202) Richness Genetic Resources of Microorganism?

Numbers of species described and currently accepted in most groups of microorganisms worldwide are respectively 143,000 & 18,500. Almost 120 new species of bacteria and 1,500 new species of fungi are added to science each year. This clearly demonstrating that knowledge of these groups is grossly inadequate.

203) Causes of changes in Gene pool?

The allele frequency in a population's gene pool can be affected by evolutionary mechanisms, such as;

• Mutation • Gene flow • Inbreeding • Natural selection, • Founder effect • Random genetic drift.

204) How antibiotic work?

“Antibiotic is chemical substance produced by microorganisms that can kill or inhibit the growth of other microorganisms”

205) Plant Genetic resources or Germplasm of a crop consist of 5 types of material?

• Land races • Obsolete varieties • Varieties in cultivation • Breeding lines • Wild form and wild relatives

206) Aquatic Genetic Resources and example?

Aquatic genetic resources include all genetic resources living in water

It include; Fish, Cyclostomes, Mussels, Decapods, Marine mammals, Aquatic plants , All other water dwelling organisms

207) The Global Strategy for Plant Conservation (GSPC)?

The Global Strategy for Plant Conservation (GSPC) is a program of the UN's Convention on Biological Diversity founded in 1999. It is a Plan to Save the World's Plant Species - grew out of the Convention on Biological Diversity and is being fed into government policy around the world.

Vision of GSPC : “Without plants, there is no life

208) Extinct species?

“A species of plant or animal that is no longer living.” e.g. Passenger Pigeon, white Rhino

209) Genetic Diversity?

“The variation in the amount of genetic information within and among individuals of a population, a species, an assemblage, or a community pakistan is the sixth largest producer of Kinow (mandarin)

210) Marco Polo sheep (Ovis ammon)

Around 200 German immigrants settled in Pennsylvania within community marriages. Developed syndrome named Ellis-van Creveld syndrome

211) Cryopreservation various methods of storage?

Cryo is Greek word. (kayos – frost). It literally means preservation in “frozen state.” Cryo-preservation or cryo-conservation is a process where organelles, cells, tissues, extracellular matrix, organs or any other biological constructs susceptible to damage caused by unregulated

Methodologies

There are various methods of storage:

1. Cryopreservation - generally involves storage in liquid nitrogen.
2. Cold storage - it involves storage in low and non-freezing temperature.
3. Low pressure – it involves partially reducing the atmospheric pressure of surrounding.
4. Low oxygen storage - it involves reducing the oxygen level but maintaining the pressure.

212) Define food security?

Conservation of plant genetic resources and animal genetic resources which are necessary for life known as food security.

213) Define sympatric speciation?

Sympatric speciation is speciation that occurs when two groups of the same species live in the same geographic location, but they evolve differently until they can no longer interbreed and are considered different species. This is often result of Reproductive isolation

214) How extinction is affected by medical study?

Many different species have unique bodily processes that can cure human diseases. E.g. the toxins produced by dart-poison frogs in the rain forest have yielded information about how alkaloid compounds behave in living organisms. Scientists also study bears for clues about how they recycle blood toxins during hibernation to find potential solutions to kidney disorders. Plants from forests are useful for medicinal purposes.

215) Why do we need to conserve plant genetic resources?

Conservation of plant genetic resources is necessary for food security and agro- biodiversity • Biodiversity provides a valuable source of compounds to the medical, food and crop protection

industries. • Maintenance of ecosystem • Genetic resources need to be conserved so that they may be used in crop research and be used as sources of genes for crop improvement.

216) Elaborate disadvantages & advantages of outbreeding?

Advantages of Out breeding:

• Out breeding often produces offspring of superior quality because it increases homozygosity (the occurrence of two alleles for the same trait at corresponding positions on homologous chromosomes) • sharply reduce the risk of deleterious recessive genes being expressed • One of the benefits of out breeding is less chance of genetic abnormalities • The ability to make a breed stronger

Disadvantages of Out breeding:

• Introduction of new genes into population • Animal discomfort: accidentally produce traits that are damaging to the health of the animal

217) Steps of conservation of PGR?

Following are steps for the Conservation of plant genetic resources

- Selection of target taxa
- Project commission
- Eco geographic survey/preliminary survey mission
- Conservation objectives
- Field exploration
- Conservation strategies
- Conserved product deposition and dissemination
- Characterization/ Evaluation
- PGR utilization
- Utilization products

218) Name the four main pillars of food?

The four pillars of food security are availability, access, utilization and stability.

219) Define migration?

Migration is a behavioural adaptation that helps animal's survival.

220) How anthropogenic activities affect migration?

- Barriers (fences, dams & skyscrapers)

- Water, air craft and fishes practices
- Telegraphic wires, towers and light houses
- Illegal hunting

221) What are land races?

A landrace is a local variety of a domesticated plant species which has developed largely adaptation to the natural and cultural environment in which it lives. It differs from a cultivar which has been selectively bred to conform to a particular standard of characteristics. Landrace populations are often variable in appearance, but they can be identified by their appearance and have a certain genetic.

222) Climate changes affect extinction?

Almost half of plant and animal species have experienced local extinctions due to climate change. Global warming could trigger not just local but global extinctions of animals and plants. Species already threatened by habitat destruction, pollution, alien invasion and overhunting are more vulnerable to climate change. Diversity of species in any one ecosystem could be affected by rises in average temperatures or a shift of climate regime

223) Describe genetic stock in plants or general populations?

Genetic stocks

Genetic stocks, broadly defined as plants or populations generated and/or selected for genetic studies, represent a unique and growing class of extremely valuable germplasm which, depending on crop, type of genetic stock and user community may represent genetic resources of either transient or long-lasting value.

Genetic stocks can be divided into three general groups:-

1. Cytological stocks
2. Mutants stock
3. Germplasm set

224) Ramsar history?

Ramsar convention came into the 21st December 1975.

225) Dolphin?

21 % out of 5,488 Mammal species and 12% out of 9,990 bird species are considered to be endangered. 4. WCPA stands for World commission on protected areas. 5. In Europe, one estimate put the value of “marketed non wood goods” from forests at €2.3 billion 6. Since the 1970, the population of the Indus Dolphin has significantly increased here.

What is botanical garden?

It is a garden dedicated to the collection, cultivation and display of a wide range of plants labelled with their botanical names. It may contain specialist plant collections such as tropical plants, or other species of plants.

226) Describe purpose of panel code?

The polluter of the environment can be punished under this code for certain types of pollution. These punishment are of following types

- Punishment for water pollution
- Punishment for atmospheric pollution
- Punishments for general pollution

227) What is location of cholistan wildlife sanctuary?

It is part of the Cholistan desert in the south eastern portion of the province of Punjab. Some of the rare animals of this region are Desert wolf (rare), Indian fox, Red fox, Jackal, Small Indian civet, Small Indian mongoose, Jungle cat and Caracal cat.

228) Define forest genetic resources?

Forest genetic resources are essential for forest-dependent communities who rely for a substantial part of their livelihoods on timber and non-timber forest products (for example fruits, gums and resins) for food security, domestic use and income generation.

229) Historical background of CBD?

Cartagena protocol was adopted on June 2001 in Cartagena, Spain. It entered into force on September 11th; 2003. Pakistan signed the Cartagena protocol in June 200. Pakistan has ratified it in May 2009.

230) Define Treaty?

Treaty, a binding formal agreement, contract, or other written instrument that establishes obligations between two or more subjects of international law”.

- Treaties do not need to follow any special form.
- A treaty often takes the form of a contract, but it may be a joint declaration or an exchange of notes.

231) Define Protected Landscape-Seascape area?

Definition: A protected area where the interaction of people and nature over time has produced an area of distinct character with significant ecological, biological, cultural and scenic value.

232) Plant genetic resources institute names?

Plant Genetic Resources Institute hosts the sole National Genebank of Pakistan for conservation of plant genetic resources and six labs including

1. Germplasm exploration lab
2. Seed preservation lab
3. in vitro conservation lab

4. Germplasm evaluation lab
5. Plant introduction and seed health lab
6. Data management lab

233) Define strict nature reserve?

Protected areas that are strictly set aside to protect biodiversity where human visitation, use and impacts are strictly controlled to ensure protection of the conservation values.

234) Role of earthworm in soil?

Earthworms help to maintain soil structure and the availability of water throughout the soil profile. Studies have shown that the presence of these animals can help to alleviate the effects of drought on crop Production. Studies have also revealed the remarkable ability of diverse soil invertebrate communities to restore the structure of degraded soil. The potential for managing soil invertebrates to enhance their beneficial roles has been little explored.

235) Second largest national park?

Kirthar National Park is the second largest national park of Pakistan spread over an area of 3000 square kilometres.

236) When and which city cartagena protocol?

Cartagena protocol was adopted on June 2001 in Cartagena, Spain. It entered into force on September 11th; 2003. Pakistan signed the Cartagena protocol in June 200. Pakistan has ratified it in May 2009.

237) Mission of Ramsar convention?

The Convention's mission is "The conservation and wise use of all wetlands through local and national actions and international cooperation, as a contribution towards achieving sustainable development throughout the world."

238) Type of gene bank?

Under suitable conditions genetic resources are conserved for a long term as gene bank. Such gene bank is of two types:

1. In vivo Gene Bank
2. In vitro Gene Bank

239) Effect of climate on invertebrates?

Most invertebrates are expected to change their geographical distribution in response to climate change so as to remain in areas to which they are well adapted. This view is strongly supported by sub-fossil evidence of insect distribution during the glaciations and interglacial periods of the Quaternary Period.

240) What is phenylketonuria?

It is a human genetic condition caused by mutations to a gene coding for a particular liver enzyme. In the absence of this enzyme, an amino acid known as phenylalanine does not get converted into the next amino acid in a biochemical pathway, and therefore too much phenylalanine passes into the blood and other tissues.

241) Define threatened species?

Threatened species are any species (including animals, plants, fungi, etc.) which are vulnerable to endangerment in the near future

242) State 2 quarantine principle?

1. When germplasm is endangered or the need for particular accessions is particularly urgent, some discretion should be possible on the part of quarantine officials in allowing exceptions for controlled entry, despite existing regulations to the contrary. 2. Decentralized quarantine services are generally more efficient because they enfold a wider range of expertise in germplasm assessment.

243) Define extinction?

“A species becomes extinct when the last existing member of that species dies.” Extinction occurs when species are diminished because of

- Environmental forces
- Evolutionary changes in their members

244) Define phenotypic plasticity?

Phenotypic plasticity is defined as the capacity of a particular genotype to express different phenotypes under different environmental conditions.

245) What are game reserves?

Definition: “A game reserve is an area wherein controlled hunting and shooting is permitted on permit basis” A game reserve (wildlife preserve) is a large area of land where wild animals live safely or are hunted in a controlled way for sport. In the game reserves the major focus is specifically the animals.

246) Define zoological park?

A zoological park is a location where animals are kept in captivity for study and viewing. Some of famous zoological parks in Pakistan are listed below,

- Bahawalpur Zoo
- Changa Manga Wildlife Park
- Clifton Fish Aquarium

- Hyderabad Zoo
- Islamabad Zoo
- Jallo Wildlife Park
- Jungle World
- Karachi Safari Park
- Karachi Zoo Walkthrough Aviary
- Lahore Zoo
- Lahore

247) Define co evolution?

Co evolution describes a situation where two or more species reciprocally affect each other's evolution. Such as the classic case of host-pathogen interaction. Further important examples of co-evolution in trees include interactions with herbivores and pollinators.

248) Who established Bahawalpur zoo?

Nawab of former state Bahawalpur, Sir Sadiq Muhammad Khan Abbasi, established Bahawalpur Zoo in 1942.

249) Germplasm conservation?

Germplasm or plant genetic resource has to be maintained in such a state that minimize the risk of loss and allows either is direct planting in the field or its preparation of planting with relative ease this is called plant genetic resources conservation or germplasm conservation.

250) Threats to FGR?

- Forests are under threat, chiefly as a result of human activities, including climate change.
- On a continent-wide scale, changes in rainfall patterns and temperature mean that some tree species will be unable to survive in their current locations. Others may be able to colonize new areas.
- Changes in climate also make it possible for pests and diseases to invade new areas, destroying the forests there.

251) Effect of climate on FGR?

These include high tree mortality through extreme climatic events, particularly drought in combination with widespread regeneration failure, for example, examined the evidence for anthropogenic climate change leading to future large scale “dieback” in Amazonian rain forest. Analysis suggested that dry-season water stress is likely to increase in eastern Amazonia over the 21st century, with the region tending toward a climate more appropriate to seasonal forests

252) What do you know about clonal repository?

Clonal repository is field Gene bank where genetic resources of clonally propagated crops like fruits are preserved as living plants. Various institutions are involved in the capacity building to develop AnGR, in the country.

253) What are Habitat species management areas?

Definition: IUCN Management Category IV (Habitat/Species Management Area) refers to areas that are managed to protect particular species or habitats. They are defined by IUCN as “protected areas aiming to protect particular species or habitats and management reflect this priority.

254) Distinguish features of protected landscape-seascape?

Distinguishing features

- Landscape and/or coastal and island seascape of high and/or distinct scenic quality and with significant associated habitats, flora and fauna and associated cultural features
- A balanced interaction between people and nature
- Unique or traditional land-use pattern

255) On how many interplaying mechanisms rely for response to environmental condition?

Tree populations rely on three interplaying mechanisms to respond to environmental change. 1. Adaptation 2. Migration 3. Phenotypic plasticity

256) What is EEZ? Exclusive Economic Zone

257) Conservation status of hingol national park?

Hingol National Park spread over an area of about 1,650 square km along the Makran Coast, Baluchistan.

258) Critical Endangered Species?

Following are the Critically Endangered Animals are:

Great Indian Bustard, Bulmer's Fruit Bat, Madagascar Pochard, Pygmy Three-Toed Sloth

259) What is cholistan protected area name?

Cholistan wildlife Sanctuary

260) For which type of material DNA Bank is useful?

DNA storage is particularly useful for those species that cannot be conserved in traditional seed or field gene banks and nor conserved in situ due to high risk in that area

261) Name physical method for storage?

- 1) Low temperature
- 2) Low light/restricted photoperiod
- 3) Minimal containment
- 4) Minimal O₂
- 5) Osmotic (water) stress

261) Legislation?

The process of making or enacting laws.

262) What is Quagga?

The quagga was a plain zebra that lived in south Africa until becoming extinct late in the 19th century. It was long thought to be a distinct species, but early genetic studies have supported it being a subspecies of plains zebra

263) Briefly explain state of world's forest genetic resources?

In 2014, the Food and Agriculture Organization of the United Nations published the first State of the World's Forest Genetic Resources. The publication addressed the conservation, management and sustainable use of forest tree.

264) What areas are covered by Cartagena protocol?

The Protocol covers Transboundary movement, transit, handling and use of all living modified organisms that may have adverse effects on the conservation and sustainable use of biological diversity, taking into account risks to human health.

265) Define seed storage?

Storage of seed is indispensable to most of plantation forestry, and the practice should not be dismissed too readily as a basic tool in maintaining genetic diversity.

266) Write the objectives of Monument features?

Primary objective: To protect specific outstanding natural features. Their associated biodiversity and habitats.

Other objectives

- To provide biodiversity protection in landscapes or seascapes that have otherwise undergone major changes
- To conserve traditional spiritual and cultural values of the site

267) Primary objectives of national park?

To protect natural biodiversity along with its underlying ecological structure other objectives. To manage the area in order to perpetuate, in as natural a state as possible. To maintain viable and ecologically functional populations. To contribute to local economies through tourism. To manage visitor use for inspirational, educational recreational purposes

268) Rann of kutch wildlife sanctuary?

It spread over 566,375 ha is part of the great Thar desert and comprises. Rann of Kutch across the frontier with India, which includes permanent saline marshes, coastal brackish lagoons, tidal mudflats, and estuarine habitats. The site supports many locally and globally threatened species, Threatened species include the Great Indian bustard, Houbara bustard, Sarus crane this area used to have the only population of the Indian Wild Ass or Onager in Pakistan.

269) Name of Botanical gardens in Pakistan?

1. Cacti and succulent plants.
2. Greenhouses, shade houses.
3. Medicinal Plants.
4. Other exotic plants.
5. Herb gardens
6. Tropical plants
7. Aromatic or textile plants

270) Plant genetic resources?

“Plant Genetic Resources for Food and Agriculture (PGRFA) are the raw material that farmers and plant breeders use to improve the quality and productivity of crops.”

271) Disadvantage of DNA bank?

There are problems with subsequent gene isolation, cloning and transfer of DNA back to a plant and it currently does not allow the regeneration of the same genotype as the original sample.

272) Function of quarantine protocol?

Quarantine practices in most countries have at least three common functions. The first is exclusion or regulatory actions to prevent or reduce the risk of entry of exotic pathogens, pests, or parasites along artificial pathways. Second is the containment, suppression, or eradication of pests or pathogens that have been recently introduced. Third is the assisting of exporters to meet the quarantine requirements of importing countries

273) Distinguishing features of habitat species management area?

- Protection of particular species
- Protection of habitats

- Active management to maintain target species
- Active management of culturally-defined ecosystems

274) Objectives of habitat and species management area?

Primary objective: To maintain, conserve and restore species and habitats.

Other objectives

- To protect vegetation patterns.
- To protect fragments of habitats as components of landscape or seascape-scale conservation strategies.
- To develop public education and appreciation of the species or habitat

275) Main objective of forest genetic resources?

- These resources are also the basis for large-scale wood production in planted forests to satisfy the worldwide need for timber and paper.
- Genetic refers to variation of genetic DNA origin, and variation of genes at different levels: 1. variation between species, 2. variation between populations within species 3. Variation between individual trees within populations.

276) Define tertiary gene pool?

Tertiary Gene Pool (GP3): The genetic material which leads to production of sterile hybrids on crossing with primary gene pool is termed as tertiary gene pool or gene pool three (GP3). It includes material which can be crossed with GP1, but the hybrids are sterile. Transfer of gene from such material to primary gene pool is possible with the help of special techniques.

277) Pakistan Terrestrial Water & Maritime Zones Act, 1976?

- It includes provisions for preservation, development and protection of marine environment
- This law controls marine pollution and exploration, development, conservation and management of living resources in Pakistan's Exclusive Economic Zone (EEZ)
- This law means that a ship carrying nuclear and hazardous substances will have to inform the Government of Pakistan.

278) What is low oxygen and low pressure storage?

For conservation of cultured plant materials low-pressure storage (LPS) and low-oxygen storage (LOS) have been developed. These are alternative methods of cryopreservation and cold storage

279) What are molecular markers?

Molecular markers are fragments of DNA which are associated with a particular region of the genome. Marker molecules can take the form of short DNA sequences, such as a sequence

surrounding a single nucleotide polymorphism, where a single base-pair change occurs. It is also known as genetic marker.

280) What are three common function of quarantine?

- a. The first is exclusion or regulatory actions to prevent or reduce the risk of entry of exotic pathogens, pests, or parasites along artificial pathways.
- b. Second is the containment, suppression, or eradication of pests or pathogens that have been recently introduced.
- c. Third is the assisting of exporters to meet the quarantine requirements of importing countries.

281) Briefly explain historical background of Ramsar convention?

Ramsar is one of the global inter-governmental environmental agreements. The treaty was negotiated in 1960s by countries and NGOs. To avoid the increasing loss and degradation of wetland habitat for migratory water birds. In an 18 nations meeting it was adopted in the Iranian city of Ramsar On 2nd February 1971. Came into force in 21st December 1975

282) Why do forest genetic resources matter?

The high levels of genetic variation that are present within many tree species can be beneficially developed and used by foresters and tree growers. Whereas agricultural crop breeders and farmers often substantially modify the growing environment to suit a specific crop species or variety, tree growers commonly identify species and provenances which can provide some improved levels of the goods and services required even without intensive selection and improvement, or intense management requirements, or major modification of the external environment.

283) Write role of pollinators?

A second major group of invertebrate providers of ecosystem services are the pollinators. It has been estimated that at least 35 percent of world food production comes from crops that are dependent on insect pollination. Pollinating insects include wild species spilling over from natural or semi-natural habitats close to crop fields, and managed pollinators (usually honey bees) that can be brought in by farmers specifically to provide pollination

284) Advantages of tissue culture?

Advantages of tissue culture conservation

- Source of disease-free material.
- It is most appropriate for rapid multiplication purposes, dissemination and active collections.

285) In vivo gene bank?

Generally plant seeds, vegetative propagules are used for storage for long time. The whole plants are preserved. This type of conservation strategy is called in vivo gene bank. In this approach, conservation method of storage is used for preservation of plant genetic resources.

286) Advantage of gene sanctuary?

1. Gene sanctuaries not only preserve the existing genetic diversity present in a population, it also allows evolution to continue. As a result new allele and gene combination would appear with time.
2. Gene sanctuary is a very good method of In situ conservation it protects the loss of genetic diversity caused by human intervention. It allows natural selection and evolution to operate.
3. It is a very good method of In Situ Conservation.

287) How FGR important?

Forests provide human beings with drinkable water, food, medicines, an environment to enjoy and fuelwood for energy, among other goods. They help to regulate the local climate. They bind soil on steep hillsides, preventing flooding and erosion further down river valleys. Globally, forests remove carbon dioxide from the atmosphere and produce oxygen.

288) Role of cryoprotectant in cryopreservation?

They are chemical which prevent cryo-destruction. These are sucrose, alcohols, glycols, some amino acid (proline), DMSO (dimethyl sulfoxide). Generally two Cryo protective agents should be used together instead of single one as they are more effective. Two common cryoprotective agents are dimethyl sulfoxide (DMSO) and glycerol. Glycerol is used primarily for cryoprotection of red blood cells, and DMSO is used for protection of most other cells and tissues.

289) List of Ramsar site in Punjab?

- Uchhali Complex
- Taunsa Barrage
- Chashma Barrage

290) Primary objective of Category VI?

Primary objective

- To protect natural ecosystems. Use natural resources sustainably, when conservation and sustainable use can be mutually beneficial.

291) Name of 3 research based botanical gardens?

- i. Karachi University Botanic Garden of Karachi University, Karachi
- Lahore Botanical Gardens, Lahore

Quaid-i-Azam University Botanical Garden, Islamabad

292) Four main mechanism of evolution?

Mutation, gene flow, genetic drift, and natural selection

293) Explain habitat loss cause of extinction?

Deforestation has killed off more species than we can count. Rainforest can disappear in the next 100 years if deforestation is not stopped. 13 million hectares of forest have been converted or destroyed Coral reefs are also threatened Reefs are home to 25% of marine animal. To date, 27% of coral reefs have been destroyed.

294) Most important reservoir for biology control agent species?

In the case of classical biological control agents, the genetic diversity of introduced populations may be relatively low because the introduction was based on a small founder population. This lack of diversity may inhibit the ability of the population to respond to climate change

295) What is the extreme weather effect on invertebrate's genetic resources?

Climate change is expected to affect all three of the main groups of invertebrate ecosystem-service providers as well as invertebrate pests. Extreme weather events such as heat waves, droughts and floods – which are predicted to increase in frequency due to climate change - are often followed by pest outbreaks. Among other contributing factors, these outbreaks can occur because the extreme event eliminates or weakens a pest's natural enemies.

296) Write the objective of Socio cultural value of national monument features?

According to IUNC there are six (6) protected area management categories that define the area according to their management. National Monument-Feature are the category III protected area. It is define as "Protected areas set aside to protect a specific natural monument."

Category III areas are likely to hold socio-cultural values as they may have features such as sacred groves, springs, waterfalls, mountains, sea coves etc. of importance to one or more faith groups. These areas are often of significant tourism value and can be managed with the objective of promoting sustainable tourism.

297) Objectives of Cartagena Protocol on Biosafety?

Objectives of Cartagena Protocol on Biosafety

The Protocol states that it aims to;

- Contribute to ensuring an adequate level of protection in the field of the safe transfer, handling and use of living modified organisms resulting from modern biotechnology that may have adverse effects on the conservation and sustainable use of biological diversity,
- It takes into account risks to human health, and specifically focusing on transboundary movements.

- It seeks to protect biodiversity from the potential risks of living modified organisms (LMOs) resulting from modern biotechnology.

298) DNA Bank?

DNA banks can now be considered as a means of complimentary conservation. DNA storage is particularly useful for those species that cannot be conserved in traditional seed or field

Advantages:

DNA banking is an efficient, simple and long-term method to conserve the genetic information.

Disadvantages:

There are problems with subsequent gene isolation, cloning and transfer of DNA back to a plant and it currently does not allow the regeneration of the same genotype as the original sample

299) Difference b/w in breeding and out breeding?

Inbreeding “The intentional breeding of distantly “Inbreeding, the mating of individuals or related or unrelated individuals for the organisms that are closely related through purpose of producing offspring of superior | common ancestry.” quality.” There are 3 types of out breeding.

There are 3 types of inbreeding 1. Cross breeding

1. Close inbreeding 2. Grading up
2. Mild inbreeding 3. Species cross
3. Line inbreeding

300) Limitation of germplasm preservation?

Seeds of many important tree species cannot adequately survive long-term storage. Seeds of many tropical hardwoods cannot survive even a year of conventional seed storage. Some genetic damage or change in gene frequencies may occur during seed storage. Cryogenic storage at its current level of technology offers great promise for long-term storage of true orthodox and sub-orthodox seeds. It has potential for application to recalcitrant species, but much more research and development will be required.

301) Link b/w botanical garden and climate change?

Plants can alter the temperature of the Earth's atmosphere. Through the process of photosynthesis, plants use energy from the sun to draw down carbon dioxide from the atmosphere and then use it to create the carbohydrates they need to grow. Since carbon dioxide is one of the most abundant greenhouse gases, the removal of the gas from the atmosphere may temper the warming of our planet as a whole. *transpiration in plants can increase water vapor in the atmosphere, causing more precipitation and cloud cover in an area. The additional cloud cover often reinforces the cooling by blocking sunlight. *Contribute to soil fertility and prevent soil erosion.

Institutes of PGR?

Plant Genetic Resources Institute hosts the sole Nation Gene bank of Pakistan for conservation of plant genetic resources and six labs including

- Germplasm exploration lab
- seed preservation lab
- in vitro conservation lab
- germplasm evaluation lab
- plant introduction and seed health lab
- Data management lab

302) Extreme weather effects on invertebrate's genetic resources?

Extreme weather events such as heat waves, droughts and floods - which are predicted to increase in frequency due to climate change - are often followed by pest outbreaks. Among other contributing factors, these outbreaks can occur because the extreme event eliminates or weakens a pest's natural enemies.

303) What are the advantages of germplasm storage?

Seeds of many valuable species can survive long term storage ('long term' is defined as spanning a period of time longer than one rotation. Good storage facilities are now available in most of the world, and they are used extensively for tree seed storage for various regeneration purposes. Seed storage is a relatively cheap method for conserving a broad range of germplasm. Large land areas are not tied up in conservation. International exchange of genetic material is facilitated by seed storage.

304) Conservation of FGR?

Safeguarding the genetic variety of forest trees is an important part of biodiversity conservation. Genetic diversity ensures the success of species in environments that are highly variable and subject to change. For most forest tree species, management plans for the conservation units allow silvicultural interventions directed towards the support of and quantity of regenerating material. Applying given critical values for the number and density of seed trees, shortening of the regeneration time, regulating competition by other tree species and controlling invasive species should also be taken into consideration. A good level of genetic diversity and reduced consanguinity in the regenerated seedlings should be a management objective. Natural regeneration should be the preferred means but if this fails to occur, assisted regeneration can be carried out using local seed lots to maintain local phenotypic identity natural regeneration both in terms of quality.

305) Write features of Cartagena protocol?

The Protocol covers: Transboundary movement, transit, handling and use of all living modified organisms that may have adverse effects on the conservation and sustainable use of biological diversity, taking into account risks to human health.

306) West Pakistan ordinance 1961?

It is an ordinance to amend and consolidate the law relating to fisheries in the West Pakistan. It says that no person will be allowed to use dynamite, pesticides or other explosives for catching the fish. Government can declare any water area as sanctuary. Then there will be no permit for fishing except some license. An Inspector of Fisheries may without a warrant arrest any person committing In his view any offence under section 6, 7, 8, 9 or 11

307) Explain method of "processing of sample" in DNA bank?

DNA preserved in DNA banks will be stored either within cells and extracted upon retrieval from storage or extracted from cells and purified before storage. The quality of the DNA is expressed through yield, purity, molecular weight, amplification efficiency and authenticity of sequences. The quality of DNA extracted from plant specimens is dependent on the condition of the specimen before storage, the storage environment and the duration of storage. Rapid drying of plant samples with silica gel or lyophilisation helps to preserve the DNA.

308) How invertebrate genetic resources cope with climate change?

Because of the many ecosystem services that they provide, invertebrates have a key role to play in adapting agriculture to the effects of climate change. The extent to which the individual services provided by invertebrates will be enhanced or impeded by climate change is difficult to predict. However, if invertebrate biodiversity is lost, the capacity of ecosystems to adapt is likely to diminish.

309) Discuss the impact of climate change on FGR. Also discuss how this changing climate effect organism which is associated with trees?

Climate change may also result in high variability in temperature and precipitation, with an increase in incidence of extreme events, such as flooding, late frosts and intensive summer droughts, amongst other events.

In some areas, such as the Mediterranean and the Neo-tropics, an increase in seasonality is also expected. Under such conditions, natural selection may not result in efficient adaptation because selection pressures are multi-directional, involving traits that may be inversely correlated at the gene level.

The standing genetic variation in populations may then not be large enough to create the rare new genotypic combinations that are required.

Ecosystems affected by abrupt change may sustain rapid and widespread transformation as ecological tipping points are exceeded.

Given the pivotal role of trees in ecosystem function, abrupt climate change impacts on them may thus have profound consequences for forests as a whole.

Irreversible loss of ecosystem integrity and function may follow, with replacement by new non endemic ecosystems.

310) Effects of changing climate on organisms associated with trees

In particular, changes in the biology of insect pests and diseases may make ecosystems more susceptible to tree mortality. Because of improved environmental conditions for the pest and reduced tree resistance due to increased stress, pests may react to climate change with range expansions and/or increases in attack severity.

311) Write the objectives of CBD?

Convention on Biological Diversity, 1992

- CBD is about the conservation and wise use of different biological resources (plants and animals). It was adopted in 1992 at Rio De Janeiro, Brazil and entered force on January, 1993, which was 90 days after the 30th ratification.
- Pakistan signed it in June 1992 at United Nations Conference on Environment Development held at Rio De Janeiro, Brazil
- Pakistan ratified it on 26th July 1994.
- The Convention on Biological Diversity covers biodiversity at all levels: 1. Ecosystems, 2. Species 3. Genetic resources
- It also covers biotechnology, including through the Cartagena Protocol on Biosafety. The three inter-related objectives are: 1. the conservation of biological diversity; 2. the sustainable use of its components; 3. the fair and equitable sharing of the benefits arising out of the utilization of genetic resources, including by appropriate
- Access to genetic resources,
- Transfer of relevant technologies,
- Funding

312) Feature of Pakistan terrestrial water and maritime act?

It includes provisions for preservation, development and protection of marine environment

- This law controls marine pollution and exploration, development, conservation and management of living resources in Pakistan's Exclusive Economic Zone (EEZ)
- This law means that a ship carrying nuclear and hazardous substances will have to inform the Government of Pakistan.

313) Enlist classes of gene pool system?

1. Primary Gene Pool (GP1)
2. Secondary Gene Pool (GP2):
3. Tertiary Gene Pool (GP3)

314) What is migration with example?

"Migration is the relatively long-distance movement of individuals, usually on a seasonal basis."
e.g. some crustaceans migrate for breeding

315) Write some feature of myrcia skeldingii?

Myrcia skeldingii was a species of plant in the Myrtaceae family. It was endemic to Jamaica. It became extinct due to habitat loss.

316) How to observe GRE interaction?

There are two main methods to analyse gene environment interaction

1. Traditional Genetic Designs 2. Molecular Analyses

317) Heterosis?

Heterosis or hybrid vigour is the superiority of the crossbred offspring. Mathematically, heterosis is the difference in performance between the crossbred and the average performance of its purebred parents.

318) Forest genetic Resources?

Forest genetic resources (FGR) are the heritable materials maintained within and among tree and other woody plant species that are of actual or potential economic, environmental, scientific or societal value.

319) What are aquatic genetic resources?

Aquatic genetic resources also comprise all water-dwelling genetic resources.

320) Which animal was listed as largest animal other than whales?

Steller's sea cow is the largest mammals, other than whales

321) Purpose of grading up?

The main purpose of the grading up is to attaining the superior traits in the breed.

322) What is gene flow?

"The introduction of genetic material (by interbreeding) from one population of a species to another"

323) When did Bubal Hart beet gone extinct?

The animals were hunted to extinction and the last known Bubal hartebeest was killed in Algeria sometime between 1945 and 1954, according to the International Union for Conservation of Nature.

324) What should be characteristic of well design strategy for national strategy for PGRFA?

- A well-designed National Strategy needs to be tailored to the particular circumstances and needs of the country,
- Should be amenable to review and updating as country situations change.

325) Tertiary Gene Pool (GP3)?

The genetic material which leads to production of sterile hybrids on crossing with primary gene pool is termed as tertiary gene pool or gene pool three (GP3). It includes material which can be crossed with GP1, but the hybrids are sterile. Transfer of gene from such material to primary gene pool is possible with the help of special techniques.

326) Ecosystem is facing massive destruction due to extinction of species how?

Species of animals cannot live away from each other. They work together to form an ecosystem. Extinction of one species can affect other by effecting food chain. They depend upon each other and their environment to survive. So in this way it can destroy whole ecosystem

327) When organism is functionally extinct?

A species may become functionally extinct when only a handful of individuals survive, which are unable to reproduce due to; • Poor health, • Age, • Sparse distribution over a large range, • A lack of individuals of both sexes (in sexually reproducing species)

328) What is the major cause of loss of genetic diversity?

- The loss of genetic diversity within a species results in the loss of useful and desirable traits (e.g. resistance to parasites).
- Reduced diversity eliminates options to use untapped resources for food production, industry, medicine.

329) Causes of migration?

- Shortage of food supply on the breeding ground
- Environmental factors

Internal factors Photoperiodism Fat deposition

330) Why do we need to conserve plant genetic resources?

Conservation of plant genetic resources is necessary for food security and agro-biodiversity. Biodiversity provides a valuable source of compounds to the medical, food and crop protection industries. Maintenance of ecosystem. Genetic resources need to be conserved so that they may be used in crop research and be used as sources of genes for crop improvement

331) Type of genetic stocks?

Genetic stocks can be divided into three general groups:

1. Cytological stocks
2. mutants stock
3. Germplasm set

332) Reason for grading up?

Breeding of animals of two different breeds where the animals of an indigenous breed/genetic group is mated by an improved pure breed for several generations towards attaining the superior traits of the improved breed. Grading up is continuous use of purebred sires of the same breed in a grade herd. By fifth generation, the graded animals may reach almost purebred levels

333) How pollution is threat to species?

Marine animals are exquisitely sensitive to traces of toxic chemicals in lakes, oceans and rivers. Drastic changes in oxygen levels, caused by industrial pollution, can suffocate entire populations. Large bodied animal and rare species are more prone to the changes caused by humans to the planet. Constant exposure to pollution can render plants and animals more susceptible to dangers including starvation, loss of habitat and disease

334) Article 6 for agricultural G. R?

The Contracting parties shall develop and maintain appropriate policy and legal measures that promote the sustainable use of plant genetic resources for food and agriculture

335) Types of out breeding?

There are 3 types of out breeding

1. Cross breeding
2. Grading up
3. Species cross

336) Huntington disease?

Huntington's disease (aka Huntington's chorea) is a genetic disorder which results in slowly progressing brain cell death. There are two distinct populations in which the disorder occurs much more often.

1. The first group is the Afrikaner population of South Africa.
2. The second group is the residents of the Lake Maracaibo region of Venezuela.

337) Difference between vertical and horizontal gene transfer?

Vertical Gene transfer

The transfer of genes from parents to offspring Transfer of genetic material occurs through sexual.

Horizontal gene transfer

Transfer of gene occur between different species Transfer of genes between organisms via methods other asexual reproduction transfer of genetic material is from parents to offspring than asexual or sexual reproduction. Genes are passed down from parent to progeny.

338) Objective of GSPC?

The GSPC highlights the importance of plants and the ecosystem services they provide for all life on earth, and aims to ensure their conservation. The GSPC has 5 main objectives:

1. Plant diversity is well understood, documented and recognized 2. Plant diversity is urgently and effectively conserved 3. Plant diversity is used in a sustainable and equitable manner 4. Plant diversity is used in a sustainable and equitable manner 5. The capacities and public engagement necessary to implement the strategy have been developed

339) Characteristics of Gene Pool?

It includes all the variants or alleles of every gene. It includes all the genes present in the population. In most cases, the population includes individuals of the same species. A gene pool includes even those genes whose effects are not visible in an individual.

Write uses of sheesham tree?

Dalbergia Sissoo, also known as Indian Rosewood, is the source of Sheesham wood. The tree is native to India and Pakistan and grows all over the Sub-Himalayan Regions. Its leaves are compound, and produces pink-white flowers that resemble a pea flower. It gives a dry fruit that is a thin and papery pale brown pod. The tree mainly offers timber

340) What is threatened of species?

“A species is likely to become endangered if it is not protected e.g. Giant Panda

Why hart's tongue fern was threatened?

Quarrying, recreation and residential development have all destroyed these plants and their habitat. Canadian populations are threatened by lumbering and the development of land for ski resorts and country estates, among other activities

341) Give diff concept of gene environment interaction?

There are two different conceptions of gene-environment interaction.

- Tabery has labelled them biometric and developmental interaction
- Sesardic uses the terms statistical and common sense interaction

342) Ratio of vegetables grown in Pakistan?

Pakistan covering 75% of the total area under vegetables, accounting for 74% of the total production. The major share in the production is of Punjab (63%) followed by Sindh (14%), Baluchistan (12%) and KPK (11%). Maximum area is grown under potatoes and about 88 % occurs in Punjab. About 46% of onion is cultivated in Sindh and 25% in Punjab. Chili is at the third position of which 84% is cultivated in Sindh

343) What is ultra-rapid freezing?

In this technique serial equilibration of embryos in high concentration of DMSO (3-5 M) supplemented with sucrose (0.3 to 0.5 M). The embryos are then plunged into liquid Nitrogen. Thawing is then done with warm water bath (approximately 500 °C per minute).

344) Importance of domestic animal resources?

Animals provide milk. Hair from goat and sheep is used for making woollen clothing, shawls and blankets some drugs are especially obtained from animals. Ex. Heparin an anti-coagulant is used to control clots in blood, is obtained from ox lungs and pig intestines. Animal's meat is the part our of diet. Animals are a great source of leather which is used for making foot wear, belts, wallets bags, furniture

345) What is horizontal gene transfer?

Horizontal gene transfer is known to occur between different species, such as between prokaryotes and eukaryotes, between the three DNA containing organelles of eukaryotes, the nucleus, the mitochondria and the chloroplast.

346) Primary genetic consequences on inbreeding?

The primary genetic consequence of inbreeding is to increase the frequency of pairing of similar genes

347) Passenger pigeons?

The passenger pigeon may have once constituted 25 to 40 percent of the bird population. The 19th century brought widespread hunting and trapping of the birds, which severely diminished their populations. The last passenger pigeon, named "Martha" died at age 29 at the Cincinnati Zoo in 1914

348) Define cryopreservation?

Cryo-preservation or cryo-conservation is a process where organelles, cells, tissues, extracellular matrix, organs or any other biological constructs susceptible to damage caused by unregulated chemical kinetics are preserved by cooling to very low temperatures (typically -80 °C using solid carbon dioxide or -196 °C using liquid nitrogen).

349) Some effect of lesser number of gene in gene population?

Populations with a lesser number of genes in their gene pool will be susceptible to problems. This may cause them to become endangered or even die altogether, i.e., become extinct

350) Why Acalypha rubrinervis known as string tree?

It was called string tree on account of the thin pendulous inflorescences which resembled red strings.

351) Write Merits of in-situ?

Plants and animals conserved in their natural environment. Biodiversity permanently protected. Natural and cultural heritage protected permanently. Ecological integrity is maintained and managed. Opportunities may arise for ecologically sustainable land uses (which come with associated economic benefits) Facilitates scientific research of the site

352) Disadvantages of inbreeding?

An increase in the prevalence of inherited disorders. A decrease in viability. A decrease in reproductive ability. The loss of genetic diversity (i.e. decrease in genetic variation). Developmental disruption, higher infant mortality and a shorter life span Reduction of immune system function.

353) Characteristics of Sigillaria?

The Sigillaria tree is one you would probably find strange today. It's a tree that doesn't reproduce via seeds, like today's deciduous and coniferous trees do. It was a spore-bearing, tree-like plant that flourished in the late Carboniferous period

354) Difference between genetic drift and natural selection?

Natural selection shows the positive changes in the genome that may give its possessor an adaptive advantage; genetic drift shows changes in the genome that may be advantageous, deleterious or may have no effect. Natural selection is response to an organism's environmental challenges. Genetic drift is completely random and is solely based on luck.

Natural selection will always result in the selection of allele that gives an advantage to its possessor, while genetic drift causes advantageous genes to be removed in the following generations.

Genetic drift is largely influenced by the population size, whereas natural selection is not.

Genetic drift leads to the reduction of genetic variations, or may sometimes be responsible for introducing genetic variation in a population. Natural selection will always result in introducing more genetic variations in a population.

355) Write a note on effect of over fishing?

The large numbers of many marine fish species and the wide ranging habitats seems virtually impossible that one fish species could be eradicated through overfishing.

Fisheries can drastically lower the numerical abundance of individual stocks or even entire fish species by overfishing For example, various cod and herring stocks in the North Atlantic

356) Difference between policy and strategy?

Policy: - A policy is a set of basic principles and guidelines, formulated and enforced by a nation government to direct and limit its actions in pursuit of long-term goals

Strategy: - A strategy refers to a concrete plan of actions, specifically designed to achieve concrete goals

357) Effect of loss in genetic diversity?

The loss of genetic diversity within a species result in the loss of useful and desirable traits (e.g. resistance to parasites). Reduced diversity eliminate options to use untapped resources for food production, industry, medicine.

358) How environment effect change gene environment interaction in case of phenylketonuria?

It is a human genetic condition caused by mutations to a gene coding for a particular liver enzyme. In the absence of this enzyme, an amino acid known as phenylalanine does not get converted into the next amino acid in a biochemical pathway, and therefore too much phenylalanine passes into the blood and other tissues. Change in environment (lowering Phenylalanine consumption) can affect the phenotype of a particular trait, demonstrating a gene-environment interaction.

359) Allopatric and sympatric speciation?

Allopatric speciation:

“Gene flow blocked by physical barriers results in Allopatric speciation”. It is geographical isolation that doesn't allow population of the same species to exchange genetic material Physical barriers to gene flow both “natural” and “artificial” Natural physical barriers include mountain ranges, oceans or vast deserts

Sympatric speciation:

Artificial physical barriers are man-made barriers such as “The Great Chine Wall barrages or dams etc Sympatric speciation is speciation that occurs when two groups of the same species live in the same geographic location, but they evolve differently until they can no longer interbreed and are considered different species. This is often result of Reproductive isolation

360) How pollution effects the aquatic organisms?

Pollution caused by industrial effluents and wastewater of private households has severely damaged the habitats of many fish species. Famous example of effect of aquatic pollution is "MINAMATA Disease." This disease was spread due to consuming mercury poisoned fishes by people

361) Gene environment example in Human?

A bunch of women from West Africa, where malarial is present, mate with a group of Europeans. Their children are less susceptible to contracting malaria due to the presence of antibodies from their West African mothers

362) Effect of bottleneck effect on allele's frequency?

Definition:

The bottleneck effect is a sharp lowering of a population's gene pool because of an environmental, or human caused, change.

Explanation:

It is an extreme example of genetic drift that happens when the size of a population is severely reduced. Events like natural disasters (earthquakes, floods, fires) can destroy a population, killing most individuals and leaving behind a small, random assortment of survivors.

Bottleneck effect and Allele Frequencies: Allele frequencies in a group may be very different from those of the population prior to the event,. Even some alleles may be missing entirely. The smaller population will also be more susceptible to the effects of genetic drift for generations (until its numbers return to normal). Effect potentially causing even more alleles to be loss.

363) How animals figure out that where they are going or, Ways of animals how they move to different areas?

Scientists aren't really sure exactly how some animals figure out how to get to where they are going. They think that:

Some animals use landmarks like rivers and streams to find their way. Some animals may navigate by the position of the sun and stars. Some animals use smell to figure out where they are going. Some species that may use the Earth's magnetic field to navigate. Example:- Some crustaceans also migrate for breeding. In many species of crabs, the females will move into shallow coastal waters to mate and lay her eggs and then they return to deeper ocean waters.

364) Brief description about animals' genetic resources for food and agriculture?

The term animal genetic resources (AnGR) is used to include all animal species, breeds and strains that are of economic, scientific and cultural interest to humankind in terms of food and agricultural production for the present or the future.

Animal genetic resources for food and agriculture:

The term "animal genetic resources for food and agriculture" is often shortened to "farm animal genetic resources" or simply "animal genetic resources" and sometimes referred to as "livestock biodiversity" or simply "livestock diversity".

Values of animal genetic resources:

- Direct use value
- Indirect use value Option value

- Bequest value
- Existence value

365) Reproduction isolation?

Reproductive isolation:

The mechanisms of reproductive isolation are a collection of evolutionary mechanisms, behaviours and physiological processes critical for speciation. They prevent members of different species from producing offspring, or ensure that any offspring are sterile. These barriers maintain the integrity of a species by reducing gene flow between related species

Type's Reproductive isolation

Pre-zygotic isolation Post-zygotic isolation

366) What is threat to AnGR?

Threats to AnGR:

Despite the importance of animal genetic resources and their diversity, their diversity has been continually decreasing over time.

One of the greatest threats to livestock diversity is pressure from large-scale commercial production systems to maintain only high-output breeds. Changes in climate will have an impact on livestock and food production in many ways.

Some major disease threats that livestock currently face include, rinderpest, foot and mouth disease, and Peste des petits ruminants (PPR), also known as sheep and goat plague.

367) Difference between gene flow and genetic Draft?

Difference Between Genetic Drift and Gene Flow:

Gene flow occurs via mixing of genes with other populations while genetic drift takes place when the allele frequency is changed between two generations of a population.

Genetic drift takes place between two generations whereas gene flow takes place between two populations Genetic drift occurs in only one species while gene flow could take place between either two populations or two species. Physical barriers matter for the gene flow but not for the genetic drift.

368) Why it is necessary to take precautionary measures to prevent the loss of genetic diversity OR Why to prevent the loss of Genetic Diversity?

The loss of genetic diversity is difficult to see or measure. In contrast, the reduction and extinction of populations is far easier to see. Extinction is not only the loss of whole species, but is also preceded by a loss of genetic diversity within the species. This loss reduces the species ability to perform its inherent role in the whole ecosystem. The loss of genetic diversity within a

species results in the loss of useful and desirable traits (e.g. resistance to parasites). Reduced diversity eliminates options to use untapped resources for food production, industry, medicine.

369) Merits of insitu and exsitu conservation?

In situ Conservation

Merits:

Plants and animals conserved in their natural environment Biodiversity permanently protected Natural and cultural heritage protected permanently Ecological integrity is maintained and managed Opportunities may arise for ecologically sustainable land uses (which come with associated economic benefits) Facilitates scientific research of the site

Ex situ Conservation

Merits:

It can be used to protect individual animals in a controlled environment. This means that issues such as predation and hunting can be monitored and managed more easily. It can be used to reintroduce species that have left an area.

370) Give different steps of Allopatric speciation?

Steps of Allopatric Speciation

1. A geographic change separates members of a population into more than one group.
2. Different gene mutations occur and build up in the different populations over time.
3. The populations become so different that members of the different populations can no longer breed with each other anymore if were they to be in the same habitat in the same time. If this is the case, allopatric speciation has occurred.

Example: Darwin finches (adaptive radiation).

371) Explain global strategy for plant conservation, or Note on GSPC?

The Global Strategy for Plant Conservation (GSPC) is a program of the UN's Convention on Biological Diversity founded in 1999. It is a Plan to Save the World's Plant Species - grew out of the Convention on Biological Diversity and is being fed into government policy around the world.

Vision of GSPC:

“Without plants, there is no life. The functioning of the planet, and our survival, depends on plants. The Strategy seeks to stop the continuing loss of plant diversity”

Objectives:

The GSPC highlights the importance of plants and the ecosystem services they provide for all life on earth, and aims to ensure their conservation. The GSPC has 5 main objectives:

1. Plant diversity is well understood, documented and recognized
2. Plant diversity is urgently and effectively conserved
3. Plant diversity is used in a sustainable and equitable manner
4. Plant diversity is used in a sustainable and equitable manner
5. The capacities and public engagement necessary to implement the strategy have been developed

372) Write a note on microbial resources?

The complexity and diversity of microbial populations is highest among all living organisms. The diversity of microbial communities and their ecologic roles are being explored in are being ulations ie

- **Soil • Water. Plants. In animals**
- In extreme environments such as the arctic deep-sea vents or high saline lakes.

Microorganisms include;

- Algae, Bacteria (including cyanobacteria), Fungi (including yeasts), Protistan groups, Viruses.

Number & Richness Genetic Resources of Microorganism:

Numbers of species described and currently accepted in most groups of microorganisms worldwide are respectively 143,000 & 18,500 120 new species of bacteria and 1,500 new species of fungi are added to science each year . This clearly demonstrating that knowledge of these groups is grossly insufficient

373) Difference between out breeding and inbreeding?

1. Out-Breeding

Out-breeding is the mating of animals of the same breed but which have no closer relationship than at least 4 to 6 generations. Outbreeding is the recommended breeding practice for most purebred sheep breeders. Within pure breeding, there are several types of mating systems. Outbreeding is the recommended breeding practice for most purebred sheep breeders.

2. In-breeding

Inbreeding is a system of breeding in which closely related animals are mated. This includes sire to daughter, son to dam, and brother to sister. Technically, inbreeding is defined as the mating of animals more closely related than the average relationship within the breed or population concerned. The primary genetic consequence of inbreeding is to increase the frequency of pairing of similar genes.

374) Plant genetic resources?

Plant Genetic Resources for Food and Agriculture (PGRFA) are the raw material that farmers and plant breeders use to improve the quality and productivity of crops. They can be defined as any genetic material of plant origin of actual or potential value for food and agriculture, e.g. seeds, tubers, mature plants etc.

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